# WHOLE-SEQUENCE WEAK CONVERGENCE OF FIXED-PHYSICAL-VOLUME LATTICE FREE FIELDS ON A PERIODIC $H^{-3}$ CARRIER

THE AIMATH PROJECT

Abstract. Constructive Euclidean field theory builds a theory from a sequence of regularised measures and must pass to a limit of those measures.

At fixed physical volume the regularisation is constrained, because a lattice of $N$ sites at spacing $a$ covers the torus only when $aN = L$. The limit is therefore genuinely infinite-dimensional, and the classical route to it runs through nuclearity of a test space and the Minlos theorem [9], which together deliver a measure on a space of distributions.

This paper takes that limit without either device. The carrier is built as an explicit weighted sequence space over the Fourier modes of the torus; a uniform moment bound yields tightness in the weaker of two negative Sobolev topologies; the finite Fourier marginals are identified in the limit; and cylinder sets generate the Borel sigma-algebra, so Borel probability measures that agree on marginals agree.

In two dimensions, for every period $L > 0$ and mass $m > 0$, the finite-source Fourier pushforwards of the lattice Gaussian laws are tight with a uniform $H^{-2}$ moment, and their finite Fourier marginals converge to paired cosine and sine Gaussian laws. The whole sequence, not merely a subsequence, then converges weakly to a Borel probability measure determined by those marginals, and the lattice covariance integrals converge to the pairing with an independently constructed periodic massive resolvent.

The theory constructed is the free (Gaussian) layer; no rate is claimed. Part II reports a Lean 4 development of the same conclusions — 89 modules, 0 sorry and 0 project axioms — and states which layer is machine-checked and which is not (Section 13).

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**How to read this paper.** Part 1 is mathematics: it fixes the lattice model, constructs the carrier, and proves that the finite-source Fourier pushforwards of the fixed-volume lattice laws converge weakly, as a *whole sequence*, to a Borel probability measure determined by their finite Fourier marginals, with every estimate proved in full. Part 2 reports the Lean 4/`Mathlib` development of the same conclusions and states precisely which layer is machine-checked and which is not. A reader interested only in the mathematics may stop at Section 10; a reader interested only in the machine verification may read Section 11 first. Structural counts quoted in the text are macros regenerated from the sources at build time, not typed values.

## Part 1. The Mathematics

### 1. Introduction

Constructing a Euclidean field theory asks for four things at once: a regularisation of the field, finite-volume laws that remain consistent as the regularisation is removed, convergence of those laws to a measure on an infinite-dimensional space, and a verification of the Osterwalder–Schrader axioms for the limit. The axioms are a fixed list [11, 12]: analyticity, reflection positivity, Euclidean invariance, clustering and regularity. Reconstruction then produces a relativistic quantum field

theory from any measure that satisfies them. The classical treatments of the programme [6, 15, 8, 10] order the four steps differently, but each of them has to pass through all four, and each of them is careful about which step is proved and which is imported. Reconstruction is the reason the fourth step matters: the measure is a Euclidean object, and the quantum field theory is recovered from it, so a theorem that stops at the measure has proved a Euclidean statement and not yet a quantum field theory.

The free (Gaussian) layer is the one place where the programme should be elementary, because the finite-volume law is Gaussian with an explicit covariance, so the limit question is a question about Gaussian laws rather than about a general family of measures. This paper settles the third step there, at fixed physical volume and in two dimensions. The torus is flat and periodic, the covariance is a function of the resolvent, and every quantity that has to converge is a sum over modes. The citations above are for position only: no step of the proof is taken from [11, 12, 9, 6, 15, 10, 8]. In particular the classical route to a measure on a space of distributions runs through the nuclearity of a test space and the Minlos theorem [9]; neither is used below, and Section 5 exhibits the concrete structure that replaces them.

The setting is fixed volume, and that is what gives the problem its shape. A lattice with $N$ sites per direction and spacing $a$ samples the torus of period $L$ exactly when $aN = L$, so the two parameters move together and the sites crowd into a fixed box rather than filling a growing one. The index of the sequence is therefore the number of sites, not the volume, and the limit is not an infinite-volume limit: the family of laws has one parameter, and the continuum it approaches is still a fixed-period torus. What is at issue is whether the sequence of Borel laws on a completed infinite-dimensional space converges, and to what.

Two statements are routinely conflated at this point. The first is convergence of the finite Fourier marginals: for each finite family of test functions the law of the resulting finite-dimensional linear image converges. This is a statement about finite-dimensional laws, it is what a lattice measurement can see, and it is a computation in the symbol domain. The second is convergence of the laws themselves, as Borel probability measures, on a completed infinite-dimensional space. The second does not follow from the first. Finite-dimensional laws can converge while no limiting Borel probability measure exists on the completed space, and a family of laws can converge on one completion and fail to converge on a weaker or stronger one. The first statement is therefore evidence for the second and not a substitute for it, and a proof that establishes only the first has not constructed a measure on the carrier at all.

Closing that gap needs three things. One needs a carrier with a Borel structure and a topology in which the assertions are made; one needs a tightness argument, since Prokhorov's theorem converts tightness into the existence of limit points and nothing else does; and one needs a proof that two Borel probability measures that agree on every cylinder coincide, which is a statement about the sigma-algebra that the cylinders generate. This paper supplies all three concretely, in the symbol and Fourier domain, with no nuclearity theorem and no Minlos theorem anywhere in the chain. The carrier is an explicit weighted sequence space, not a nuclear test space; tightness is a single uniform moment and a compactness comparison of two weights; cylinder determination is proved from separability of that space.

The argument has five moves, in this order.

(i) Fix the model and the covariance limit (Section 3, Section 4). The lattice symbol is computed from the discrete Laplacian, the physical-volume Gaussian law is defined, and the covariance integrals are proved to converge against a periodic massive resolvent that is constructed independently of the Fourier series which is then shown to equal it. The lattice covariance itself is an input object (Remark 3.5); what is proved is the limit of its pairings, for bounded continuous periodic tests.

(ii) Build the carrier (Section 5). The completion is realised as an explicit weighted $\ell^2$ space over the mode lattice, so separability and the equality of the cylinder sigma-algebra with the

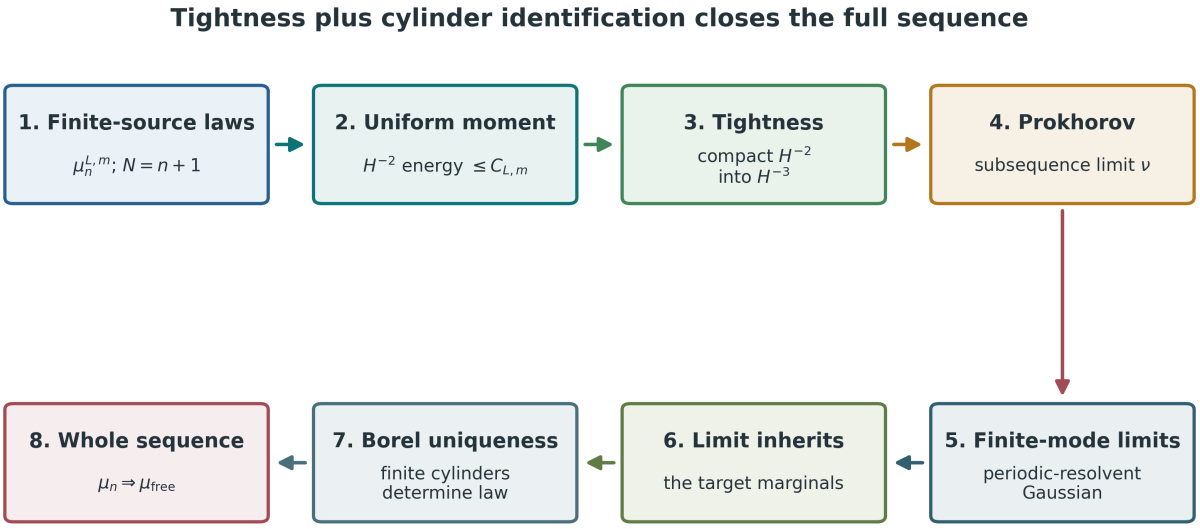



FIGURE 1. The spine of the limit argument. The boxes are numbered in the order the proof visits them: the finite-source lattice laws; the uniform $H^{-2}$ moment; tightness on the carrier; Prokhorov extraction of a subsequential limit; the finite-mode limits from the marginal computations; the limit inherits the target marginals; cylinder-to-Borel uniqueness identifies that limit; and whole-sequence convergence closes it. The last step is the whole-sequence conclusion, not merely a subsequential one. Two inputs are not drawn, because they are settled before this half begins: the carrier construction (move 2 of Section 1) and the covariance limit (part of move 1), which is where the target covariance comes from. The figure is drawn by `figures/generate_spine.py` from the same notation this paper uses, so its footer states the boundary of Section 9 rather than a paraphrase of it.

Borel sigma-algebra are theorems about a concrete space rather than hypotheses attached to an abstract one, and the coordinate maps are continuous and injective (Lemma 5.5).

(iii) Prove one moment bound and deduce tightness (Section 6). A uniform $H^{-2}$ moment with an explicit constant makes the Markov tail small, and sublevel sets of the $H^{-2}$ energy are relatively compact in the weaker $H^{-3}$ topology, which is exactly what Prokhorov's theorem needs.

(iv) Identify the finite Fourier marginals (Section 7). For each finite family of modes the law of the corresponding finite collection of coordinates converges to the paired cosine/sine Gaussian law whose covariance is the resolvent pairing of step 1.

(v) Identify the limit and pass to the whole sequence (Section 8, Section 9). Every subsequential limit inherits those marginals by continuity of the cylinder maps; cylinder uniqueness gives one limit for all of them; and the subsequence criterion upgrades subsequential convergence to convergence of the sequence.

The figure shows the spine of the argument, from the lattice laws through the moment bound and tightness to the whole-sequence conclusion, in the order in which the steps are proved.

What makes the argument checkable rather than merely correct is the following.

- The covariance limit (Theorem 4.4) is proved against the resolvent of Lemma 4.1, which is built from an energy identity and a Riesz adjoint and is therefore available before any Fourier series is written. The symbol is an output of that construction, not a postulate that the lattice limit is then matched to. The limit of the sampled covariance is therefore settled independently of what the series says it is.
- The carrier is the explicit weighted $\ell^2$ space of Definition 5.1. Separability (Theorem 5.6) and the equality of cylinder and Borel sigma-algebras (Theorem 5.7) are proved for that space, so the two structural facts that a measure-theoretic limit normally assumes are theorems here.
- The conclusion (Theorem 9.1) is about the whole sequence, and Corollary 9.3 shows that no subsequence can produce a different limit. The two quantitative inputs are the uniform moment (Theorem 6.2) and the marginal limits (Theorem 7.4); the identification step rests on Theorem 8.1.
- The target law of each finite family is computed and not selected. Its characteristic function is derived from the resolvent pairing of step 1 (Theorem 7.2) and its covariance entries are matched to that pairing, so the marginal limits are a consequence of the covariance limit rather than a second, independent assumption about what the limit should be.

Two items are stated in the underlying documents without a paper-level derivation, and are derived in full here: the uniform trace constant behind the moment bound (Lemma 6.1) and the entrywise identification of the paired cosine/sine covariance matrix (Lemma 7.3). Both are elementary once the normalisation of the lattice covariance is fixed, and both are needed before the tightness and marginal steps can be completed.

The scope of the result is deliberately narrow. What converges is the finite-source Fourier pushforward $\mu_N$ of the lattice Gaussian law, which Lemma 5.8 identifies as the law of a Borel carrier-valued map of the lattice field, so "the lattice field laws converge" is asserted in exactly that sense and no other: the statement is about the law of the Fourier coefficient sequence, not about a particular total lift of the development (Remark 5.9). No rate of convergence, no exponential tail and no quantitative modulus is claimed: the tightness argument is Markov's inequality applied to a second moment of the norm, and nothing else. No interacting ultraviolet limit and no Osterwalder–Schrader reconstruction is proved; the theory constructed here is the free one. The carrier, its weights and every object in the measure limit are two-dimensional, and no dimensional generalisation of them is claimed. Section 10 states the boundary in full and Section 13 states which layer of the argument is machine-checked.

## 2. Setting

This section fixes the objects and the conventions that the rest of the paper uses, and it proves nothing substantive. Definition 2.1 fixes the torus, its test space and the Fourier coefficient; Remark 2.2 the normalisation of the pairings; Definitions 2.3 and 2.4 the index sets and the three weights; Definition 2.5 the negative-Sobolev sequence spaces that carry the measures and the energy. Hypothesis 2.6 fixes the fixed-volume regime and the dictionary between the two ways of indexing it. Two background facts are stated: one is borrowed, as an external theorem (external theorem 2.7), and the other is proved outright below (Lemma 2.8). No probabilistic object is introduced here: the lattice laws belong to Section 3.

### 2.1. **The torus, its test space, and the Fourier coefficient.**

**Definition 2.1** (The periodic torus and its test space)**.** Let $L > 0$. The *periodic torus* is $T_L^2 := \mathbb{R}^2/(L\mathbb{Z})^2$, identified throughout with the half-open cell $[0, L)^2$; its volume is $V := L^2$. A *test function* is a bounded continuous $f : \mathbb{R}^2 \to \mathbb{R}$ that is $L$-periodic in each coordinate,

$$f(x + Le_i) = f(x) \qquad \text{for } i = 1, 2 \text{ and all } x \in \mathbb{R}^2,$$

where $e_1, e_2$ are the standard unit vectors, so a test function is a bounded continuous real function on $T_L^2$. The trigonometric polynomials are uniformly dense in the set of test functions — a standard fact, recorded here for orientation; no step below spends it, because every approximation in this paper is a finite-mode truncation of a series whose terms are already explicit. The *Fourier coefficient* of a test function $f$ at the mode $k \in \mathbb{Z}^2$ is

$$F(k) := \int_{[0,L]^2} f(x)\, e^{-2\pi i\, k\cdot x/L}\, \mathrm{d}x, \tag{1}$$

the integral being taken over the physical cell of Lebesgue measure $V$, so that $F(k)$ carries the dimension of $f$ times volume. There is deliberately no factor $V^{-1}$ in (1).

*Remark* 2.2 (The pairings carry $V^{-1}$)*.* The coefficient (1) is the *integral-normalised* one. Every Fourier pairing written later has the shape $V^{-1}\sum_k \overline{F(k)}\, G(k)\, \rho(k)$ with a symbol $\rho$: the factor $V^{-1}$ belongs to the pairing, never to the coefficient. The competing convention is the orthonormal one attached to the basis of external theorem 2.7, whose coefficient at $k$ is $V^{-1}F(k)$; the change of variables between the two is Lemma 3.7. Both conventions occur below, and interchanging them loses exactly one factor of $V = L^2$.

### 2.2. **Modes, weights, and which norm enters them.**

**Definition 2.3** (Fourier index set and weights)**.** The *continuum index set* is $\mathbb{Z}^2$; the *lattice index set* is $I_N := (\mathbb{Z}/N\mathbb{Z})^2$, with $N \in \mathbb{N}$ as in Hypothesis 2.6. For $k \in \mathbb{Z}^2$ put $\|k\|_\infty := \max\{|k_1|, |k_2|\}$. The weights of this paper are functions of this sup norm and of nothing else; the Euclidean norm enters only the symbols (7)–(8). The weights are

$$w_3(k) = \begin{cases} 1, & k = 0, \\ \|k\|_\infty^{-6}, & k \neq 0, \end{cases} \qquad w_2(k) = \begin{cases} 0, & k = 0, \\ \|k\|_\infty^{-4}, & k \neq 0, \end{cases} \tag{2}$$

together with the *full* weight $\widetilde{w}_2$, which agrees with $w_2$ at every $k \neq 0$ and takes the value $\widetilde{w}_2(0) = 1$. Thus $w_3$ and $\widetilde{w}_2$ are positive at every mode, the origin included, whereas $w_2$ is not.

The three weights are not interchangeable, and everything below depends on keeping them apart. The seminorm weight $w_2$ annihilates the zero mode and enters only where that mode is restored by a separate additive term — the split that makes the weighted trace finite (Theorem 6.2 and Lemma 6.1). The full weight $\widetilde{w}_2$ is positive everywhere, so the weighted sum it defines is an honest norm and its sublevel sets are the compact sets of the argument (Theorem 6.4). The weight $w_3$ is positive everywhere for the same reason as $\widetilde{w}_2$, and its weighted space carries the measures (Definition 2.5).

**Definition 2.4** (The sup norm on the mode lattice)**.** For $k \in \mathbb{Z}^2$ the symbol $\|k\|_\infty$ denotes the sup norm $\max_i |k_i|$ of (2), and $|k|$ the Euclidean norm $(\sum_i k_i^2)^{1/2}$. Every weight of this paper is a function of $\|k\|_\infty$ alone, every symbol of Eqs. (7) and (8) a function of $|k|$. The two are equivalent, $\|k\|_\infty \leq |k| \leq 2^{1/2}\, \|k\|_\infty$ in two dimensions, and they give the same summability threshold (Lemma 2.8); they are not interchangeable inside an estimate, because for $k$ with $|k_1| \neq |k_2|$ the two differ and $\|k\|_\infty^{-4} \neq |k|^{-4}$.

2.3. **Negative-Sobolev carriers.**

**Definition 2.5** (Negative-Sobolev sequence spaces)**.** Let $\mathcal{W} = (\mathcal{W}(k))_{k\in\mathbb{Z}^2}$ be a weight with $\mathcal{W}(k) > 0$ for every $k$, and put

$$\ell^2(\mathbb{Z}^2;\mathcal{W}) := \Big\{ c = (c_k)_{k\in\mathbb{Z}^2} \in \mathbb{C}^{\mathbb{Z}^2} : \sum_{k\in\mathbb{Z}^2} \mathcal{W}(k)\,|c_k|^2 < \infty \Big\}, \tag{3}$$

a Hilbert space for the inner product $\langle c, d\rangle_{\mathcal{W}} = \sum_k \mathcal{W}(k)\overline{c_k}d_k$. For $s > 0$ let

$$\widetilde{w}_s(k) = \begin{cases} 1, & k = 0, \\ \|k\|_\infty^{-2s}, & k \neq 0, \end{cases} \qquad \text{and set} \qquad H^{-s} := \ell^2(\mathbb{Z}^2;\widetilde{w}_s). \tag{4}$$

The two exponents of (2) are the cases $\widetilde{w}_2$ and $\widetilde{w}_3 = w_3$ of this family. This is a definition and not a theorem: no separately defined distribution space is identified here, in keeping with Section 10. The letter $s$ is the Sobolev order, so the weight of $H^{-s}$ decays like $\|k\|_\infty^{-2s}$; where the *decay exponent* itself is meant below, it is written out rather than denoted by $s$, because the two differ by a factor of two and the summability threshold is stated in the decay exponent.

Two of these spaces occur, with two different jobs, and neither exponent may be relabelled. The *carrier* is $H^{-3}$: the lattice laws of Section 3 are Borel probability measures on its completion, and the weak convergence asserted in Section 9 is weak convergence there. The *energy* is $H^{-2}$: the uniform moment of Theorem 6.2 is an $H^{-2}$ quantity. The inclusion $H^{-2} \subseteq H^{-3}$ is Proposition 5.4, and it is the ratio $w_3/\widetilde{w}_2 = \|k\|_\infty^{-2}$ that tends to zero — the $H^{-3}$ weight decays *faster*, not the $H^{-2}$ one — which is what makes an $H^{-2}$-sublevel set have small $H^{-3}$-tails, hence relatively compact closure (Theorem 6.4).

A point of $H^{-3}$ may be written as its family of Fourier coefficients, with the weighted norm (4), or as the family of *stored coordinates* obtained by applying $\sqrt{\widetilde{w}_3(k)}$ to each coefficient; the two differ by the diagonal isometry of Definition 5.1, and Remark 5.2 says which one each formula below is written in. The deweighting is defined on all of $H^{-3}$ precisely because $\widetilde{w}_3(k) > 0$ at every $k$, the zero mode included (Eq. (2)) — which is why the zero mode of the $H^{-3}$ weight is 1 and not 0. Computing an $H^{-2}$ norm in coordinates that already carry the $H^{-3}$ weight multiplies it by a mode-dependent factor, which is the trap Remark 2.10 warns about.

2.4. **Fixed volume, the two symbols, and a borrowed fact.**

**Hypothesis 2.6** (Fixed physical volume)**.** Throughout, $L > 0$ and the mass $m > 0$ are fixed. The lattice spacing $a > 0$ and the lattice size $N \in \mathbb{N}$ satisfy

$$aN = L, \tag{5}$$

and the sequence of lattices is indexed by $n \in \mathbb{N}$ through

$$N = n + 1, \qquad a = a_n = \frac{L}{n+1}\ . \tag{6}$$

Hypothesis 2.6 also fixes the bookkeeping that the two halves of the paper index differently. The mathematics below writes $N$ for the lattice size and passes to $N \to \infty$; the machine development of Section 11 indexes its statements by $n$ and derives $N = n + 1$, $a = L/(n + 1)$. Equation (6) is the whole dictionary, with one consequence: the mesh is never a free parameter, so only $L$, $m$ and $n$ are free below.

Fixed volume buys one thing, and the argument rests on it: a symbol limit. Because (5) forces $a = L/N \to 0$ at fixed $L$, the lattice symbol below approaches the continuum symbol mode by mode at each fixed $k$, and it is exactly because the spacing vanishes while $L$ is held fixed that the limit is not a rescaling of the field. Fixed volume buys no modulus: no rate, no exponential tail and no quantitative convergence statement is proved here (Section 10), and the limit is not asserted to be an $L^2$ function.

Two symbols are used from here on, and neither identity is proved here: the first is the symbol of the operator of Definition 3.1 and Proposition 3.2, the second that of the resolvent of Lemma 4.1. The lattice configuration space is $\mathbb{R}^{I_N}$: a configuration assigns a real number to each site $x \in I_N$, and the site $x$ has the physical representative $ax \in [0, L)^2$. The lattice symbol and the continuum symbol are

$$\lambda_N(p) := m^2 + 4a^{-2} \sum_{i=1}^{2} \sin^2\Big(\frac{\pi p_i}{N}\Big), \qquad p \in I_N, \tag{7}$$

$$\lambda(k) := m^2 + \Big(\frac{2\pi}{L}\Big)^2 |k|^2, \qquad k \in \mathbb{Z}^2. \tag{8}$$

The expression (7) is well defined on the residue class $p_i \in \mathbb{Z}/N\mathbb{Z}$ because $\sin^2(\pi t)$ is 1-periodic in $t$. Both symbols are real and at least $m^2 > 0$, so no denominator formed from them vanishes.

**External theorem 2.7** (Torus Fourier basis)**.** Let $z = x/L \in [0,1)^2$ and let $\mathrm{d}z$ be Lebesgue measure on the unit cell, whose total mass is 1. The family $z \mapsto e^{2\pi i\, k\cdot z}$, $k \in \mathbb{Z}^2$, is a complete orthonormal basis of $L^2([0,1)^2, \mathrm{d}z; \mathbb{C})$.

External theorem 2.7 is used as a black box, and no other completeness statement is needed below. It is quoted in this orthonormal form because that is the form in which tests are paired with the resolvent in Section 4; the factor that reconciles it with (1) is the $V^{-1}$ of Remark 2.2, whose proof is Lemma 3.7.

**Lemma 2.8** (Summability of the weights)**.** *In two dimensions both of the following series converge:*

$$\sum_{k\neq 0} \|k\|_\infty^{-4} < \infty, \qquad \sum_{k\neq 0} \|k\|_\infty^{-6} < \infty. \tag{9}$$

*Proof.* The two sums are explicit. The shell $\|k\|_\infty = q$, $q \geq 1$, carries $(2q+1)^2 - (2q-1)^2 = 8q$ modes, and on that shell $\|k\|_\infty^{-r} = q^{-r}$, so each sum is $8\sum_{q\geq 1} q^{1-r}$ with $r = 4$ respectively $r = 6$; such a series converges exactly when $r > 2$. The inequality that matters is therefore $2 < 4$ and $2 < 6$: *the decay exponent of the weight must exceed the dimension.* □

The two values are worth naming, because the constants below carry them:

$$\sum_{k\neq 0} \|k\|_\infty^{-4} = 8\zeta(3), \qquad \sum_{k\neq 0} \|k\|_\infty^{-6} = 8\zeta(5).$$

*Remark* 2.9 (The exponents are the two-dimensional ones). The threshold just computed is that the decay exponent exceeds $d$, and here $d = 2$. In $d = 4$ the decay exponent 4 of $\widetilde{w}_2$ would be critical rather than summable, so the pair $w_2, w_3$ of (2) is a two-dimensional pair; the paper claims nothing about dimension four and offers no general one. The source of Section 4 is registered for a general dimension $d$, but this paper writes it in two dimensions, like everything else here: the test functions are functions on $\mathbb{R}^2$, and the shell count in the last step of its proof uses $d = 2$. The measure chain therefore uses $d = 2$ throughout, with no statement and no object of this paper claimed in another dimension (Section 10).

*Remark* 2.10 (Four conventions a reader can get wrong). Each of the following is settled above, and each is easy to reverse by accident. *(i) Integral versus orthonormal normalisation:* (1) carries no $V^{-1}$, the pairing carries it, and the probability-torus form appears only where a test is paired with the resolvent in Section 4 (Remark 2.2 and Lemma 3.7). *(ii) Sup norm versus Euclidean norm:* the weights are stated for $\|k\|_\infty$ and the symbols for $|k|$, and the two must not be exchanged inside an estimate (Definition 2.4). *(iii) Seminorm versus full weight:* $w_2$ vanishes at the zero mode and is usable only where that mode is supplied separately, while $\widetilde{w}_2$ and $w_3$ are positive everywhere and are what the norm and the compactness argument use (Definitions 2.3 and 2.5). *(iv) Carrier*

*versus energy exponent:* $H^{-3}$ is the carrier of the laws and the topology of weak convergence, $H^{-2}$ the exponent of the energy and of the uniform moment, and neither may stand in for the other (Definition 2.5, Theorem 6.2, and Section 6).

## 3. The lattice free field and its physical-volume law

This section fixes the two objects that Part I consumes. Definition 3.1 introduces the discrete Laplacian and the Gaussian operator; Proposition 3.2 proves the eigenfunction identity and the positivity bound that every later estimate uses. Lemma 3.3 is the one estimate of the paper that spends boundedness, Definition 3.4 is the physical lattice Gaussian law, and Definition 3.6 carries that law to the weighted Fourier side. Lemma 3.7 reconciles the two Fourier normalisations of Remark 2.2. Two points made in passing matter later: the lattice covariance is an input and not derived here (Remark 3.5), and the spacing enters more than once, in different powers (Definition 3.4).

### 3.1. The discrete Laplacian and the Gaussian operator.

**Definition 3.1** (The discrete Laplacian and the Gaussian operator)**.** Let $N \geq 1$ and $a = L/N$ as in Hypothesis 2.6. A *configuration* is a real function $\omega$ on the lattice $I_N = (\mathbb{Z}/N\mathbb{Z})^2$ of Definition 2.3; the lattice configuration space is $\mathbb{R}^{I_N}$, and we also admit complex $u : I_N \to \mathbb{C}$. With $e_1, e_2$ the unit residues, the *discrete Laplacian* is

$$(\Delta_a u)(x) := a^{-2} \sum_{i=1}^{2} \big( u(x + e_i) + u(x - e_i) - 2u(x) \big), \tag{10}$$

the two oriented neighbours being subtracted and added exactly once, so that (10) keeps its form even when $N \leq 2$. The *Gaussian operator* is

$$Q := m^2 - \Delta_a, \tag{11}$$

and we write $A_N := Q$ when $Q$ acts on configurations against the *counting* inner product, the one attached to the vector space $\mathbb{R}^{I_N}$:

$$\langle u, v \rangle_{\text{counting}} := \sum_{x \in I_N} u(x) v(x), \tag{12}$$

so that $A_N^{-1}$ is the inverse of $Q$. For $p \in I_N$ the *character* at $p$ is the function

$$\chi_p(x) := \exp\big( 2\pi i \, p \cdot x / N \big), \qquad x \in I_N,$$

the product $p \cdot x$ being read modulo $N$.

**Proposition 3.2** (Operator facts)**.** *$\Delta_a$ is a symmetric real matrix, hence self-adjoint, and $Q$ is positive definite with spectrum contained in $[m^2, \infty)$. Consequently $Q$ is invertible and has a positive square root $Q^{1/2}$, and $\mathbb{C}^{I_N}$ has an orthonormal basis of eigenvectors of $Q$. The eigenfunctions are the characters, with eigenvalues the symbol $\lambda_N$ of* (7)*:*

$$\Delta_a \chi_p = -4a^{-2} \Big( \sum_{i=1}^{2} \sin^2 \frac{\pi p_i}{N} \Big) \chi_p, \qquad Q\chi_p = \lambda_N(p) \, \chi_p, \tag{13}$$

*and $\lambda_N(p) \geq m^2 > 0$ for every $p \in I_N$.*

*Proof. The eigenfunction identity.* Translating the argument of a character multiplies it by a scalar: for $i = 1, 2$ and every $x$,

$$\chi_p(x + e_i) = \zeta_i \chi_p(x), \qquad \chi_p(x - e_i) = \zeta_i^{-1} \chi_p(x), \qquad \zeta_i := e^{2\pi i p_i / N},$$

because $p_i(x_i + 1) \equiv p_i x_i + p_i$ modulo $N$. Inserting this into (10) gives $(\Delta_a \chi_p)(x) = a^{-2} \sum_i (\zeta_i + \zeta_i^{-1} - 2) \chi_p(x)$, and Euler's formula gives $\zeta_i + \zeta_i^{-1} - 2 = 2\cos(2\pi p_i / N) - 2 = -4 \sin^2(\pi p_i / N)$. Hence

the eigenfunction identity of (13); the relation $Q\chi_p = \lambda_N(p)\chi_p$ then follows from the definition (7) of the symbol together with $Q = m^2 - \Delta_a$.

*Positivity.* Each summand $4a^{-2}\sin^2(\pi p_i/N)$ of (7) is a nonnegative real number, so $\lambda_N(p) \geq m^2$; and $m > 0$ by Hypothesis 2.6.

*An explicit orthonormal eigenbasis.* Summing the one-coordinate geometric series and using $\zeta^N = 1$ for $\zeta = e^{2\pi i/N}$,

$$\sum_{x\in I_N} \chi_{-p}(x)\chi_q(x) = \prod_{i=1}^{2} \sum_{x_i\in\mathbb{Z}/N\mathbb{Z}} e^{2\pi i(q_i-p_i)x_i/N} = N^2\mathbf{1}_{p=q},$$

so the $N^2$ characters are pairwise orthogonal and nonzero, hence a basis of $\mathbb{C}^{I_N}$; scaled by $N^{-1}$ they are orthonormal, and (13) makes each of them an eigenvector.

*Self-adjointness and the spectrum.* The coefficient of $u(y)$ in $(\Delta_a u)(x)$ is $-2a^{-2}$ when $y = x$, $a^{-2}$ when $y - x = \pm e_i$ for one $i$, and 0 otherwise; this is symmetric in $x$ and $y$, so $\Delta_a$ is a symmetric real matrix, hence self-adjoint, and $Q$ is too. On the basis just exhibited $Q$ acts diagonally with the real entries $\lambda_N(p)$, so its spectrum is exactly $\{\lambda_N(p) : p \in I_N\} \subseteq [m^2, \infty)$: the operator is positive definite, hence invertible. A positive definite symmetric matrix has a positive definite symmetric square root, and that is the finite-dimensional spectral theorem, the only fact named in this proof; the eigenbasis itself is exhibited above rather than postulated. Write $Q^{1/2}$ for that root. □

**Lemma 3.3** (Finite Parseval)**.** *Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be bounded, and for $p \in I_N$ let*

$$F_N(p) := a^2 \sum_{x\in I_N} f(ax)\,\chi_{-p}(x), \qquad G_N(p) := a^2 \sum_{x\in I_N} g(ax)\,\chi_{-p}(x) \tag{14}$$

*be the sampled coefficients at the physical points $ax$ of Hypothesis 2.6, the factor $a^2$ being the area of one cell. Then*

$$\sum_{p\in I_N} \overline{F_N(p)}\,G_N(p) = Va^2 \sum_{x\in I_N} f(ax)g(ax), \qquad \sum_{p\in I_N} |F_N(p)|^2 = Va^2 \sum_{x\in I_N} |f(ax)|^2 \leq V^2 \|f\|_\infty^2 . \tag{15}$$

*Proof.* Summing the one-coordinate geometric series over modes instead of over sites gives the dual orthogonality relation $\sum_{p\in I_N} \chi_p(x)\chi_{-p}(y) = N^2$ when $x = y$ and 0 otherwise. Since $f$ is real, $\overline{\chi_{-p}(x)} = \chi_p(x)$; inserting (14) and interchanging the finite sums therefore collapses the cross terms onto the diagonal:

$$\sum_{p\in I_N} \overline{F_N(p)}\,G_N(p) = a^4 \sum_{x,y\in I_N} f(ax)g(ay) \sum_{p\in I_N} \chi_p(x)\chi_{-p}(y) = a^4N^2 \sum_{x\in I_N} f(ax)g(ax).$$

The bilinear identity of (15) follows from $a^2N^2 = a^2 \cdot L^2/a^2 = V$; the diagonal one is the case $g = f$, and $a^2 \sum_x |f(ax)|^2 \leq a^2N^2 \|f\|_\infty^2 = V \|f\|_\infty^2$. □

This is the only place in Part I where a test is spent through boundedness rather than through smoothness: the bound is $\|f\|_\infty$ alone, and that is what lets Section 4 work with bounded continuous periodic tests and ask for no derivative.

### 3.2. **The physical lattice Gaussian law.**

**Definition 3.4** (The physical lattice Gaussian law)**.** With $A_N = Q$ of (11), put

$$C_N := a^{-1}A_N^{-1/2}, \qquad \text{so that} \qquad C_N^2 = a^{-2}A_N^{-1}; \tag{16}$$

the squared object $C_N^2$ is the covariance operator. The *physical lattice Gaussian law* $\mu_N^{\mathrm{GJ}}$ is the centred Gaussian Borel probability measure on $\mathbb{R}^{I_N}$ that is the law of $C_N\xi$ for a standard Gaussian vector $\xi$ on $\mathbb{R}^{I_N}$, so that $C_N^2$ is its covariance operator. Its test averages are $\omega \mapsto \omega(f) :=$

$\sum_{x\in I_N} a^2 f(ax)\,\omega(x)$, so that $\omega(f) = \langle b, \omega\rangle_{\text{counting}}$ with the sampled vector $b_x := a^2 f(ax)$, $x \in I_N$. For $b, c \in \mathbb{R}^{I_N}$ the covariance pairing is therefore

$$\langle C_N b, C_N c\rangle = a^{-2}\,\big\langle b, A_N^{-1} c\big\rangle_{\text{counting}}\,. \tag{17}$$

It is the factor $C_N$, not the operator $C_N^2$, that the rest of Part I calls the lattice covariance.

Only two properties of $\mu_N^{\mathrm{GJ}}$ are used later, and they are its operative content here. The mean is exactly zero: $\int \omega(f)\,\mu_N^{\mathrm{GJ}}(\mathrm{d}\omega) = 0$ for every bounded $f$, because $\omega(f)$ is linear in $\omega$ and the law is centred. The covariance is the inverse-mass identity (17) in sampled form: with $b_x = a^2 f(ax)$ and $c_x = a^2 g(ax)$,

$$\mathrm{Cov}\big(\omega(f), \omega(g)\big) = \langle C_N b, C_N c\rangle = a^{-2}\,\big\langle b, A_N^{-1} c\big\rangle_{\text{counting}} = a^2\,\big\langle f, A_N^{-1} g\big\rangle_{\text{counting}}\,, \tag{18}$$

the last equality because the counting sum picks up one factor $a^2$ per sampled vector. No other property of $\mu_N^{\mathrm{GJ}}$ enters any estimate below.

Where the powers of $a$ sit is worth stating once, explicitly, because the factors pull in opposite directions and a reader who cancels them gets a constant wrong. There are three occurrences and they have three different jobs. *(i)* The cell area $a^2$ sits on the *test evaluation*: it is the quadrature weight of the sum $\omega(f) = \sum_x a^2 f(ax)\omega(x)$, hence part of the sampled vector $b_x = a^2 f(ax)$ in (18). *(ii)* The covariance factor carries $(a^2)^{-1/2} = a^{-1}$, and the pairing (17) squares it, so $a^{-2}$ sits on the bilinear form. *(iii)* The same $a^2$ reappears inside the per-site vector of the lift (19), where it is again the cell area and is again a quadrature weight, this time a Fourier one. The cancellation between *(i)* and *(ii)* is exact and leaves the single factor $a^2$ of (18), since $a^{-2}(a^2)^2 = a^2$. The $a^2$ of *(iii)* is not the inverse of the $a^{-2}$ of *(ii)*: the two belong to different functionals, and neither may be cancelled against the other.

*Remark* 3.5 (The lattice covariance is an input). $C_N$ is data: (16) fixes it, and Proposition 3.2 is what makes that specification consistent, since $Q$ is then invertible with a positive square root. It is not derived here from any field-theoretic construction, and no property of $C_N$ beyond exact zero mean and (17) is used anywhere below. The name GJ attached to this normalisation records which power of the spacing is meant — namely $a^{-1}$ per factor in (16) — and it is a name and not a citation: no step of the proof is taken from a reference.

### 3.3. **The finite-source Fourier lift and its induced law.**

**Definition 3.6** (The finite-source Fourier lift)**.** For a configuration $\omega \in \mathbb{R}^{I_N}$ and a mode $p \in \mathbb{Z}^2$ put

$$(L_N\omega)(p) := \sqrt{w_3(p)}\; a^2 \sum_{x\in I_N} \omega(x)\, e^{-2\pi i\, p\cdot(ax)/L}, \tag{19}$$

with $w_3$ the weight of Definition 2.3, and let

$$\mu_N := (L_N)_*\,\mu_N^{\mathrm{GJ}} \tag{20}$$

be the image of the law of Definition 3.4 under $L_N$.

The phase in (19) is the physical one, $e^{-2\pi i\, p\cdot(ax)/L}$, and since $a/L = 1/N$ it equals $\chi_{-p}(x)$: the lift is the discrete Fourier transform of the sampled field, not of the residue index, and that is what makes $a^2$ the cell area rather than a bookkeeping symbol. Because the phase has period $N$ in each component of $p$, the family (19) is the $N^2$-periodic repetition over $\mathbb{Z}^2$ of its values on $I_N$. Each entry is a finite real-linear combination of the coordinate evaluations $\omega \mapsto \omega(x)$, so $L_N$ is Borel as a map into $\mathbb{C}^{\mathbb{Z}^2}$; and $L_N$ lands in the carrier of Definition 2.5, not merely in some larger space, because with $\|\omega\|_1 := \sum_x |\omega(x)|$,

$$\sum_{p\in\mathbb{Z}^2} |(L_N\omega)(p)|^2 \le a^4\,\|\omega\|_1^2 \sum_{p\in\mathbb{Z}^2} w_3(p) = a^4\,\|\omega\|_1^2 \Big(1 + \sum_{p\neq 0} \|p\|_\infty^{-6}\Big) < \infty$$

by Lemma 2.8, and $w_3 \le 1$ makes every weighted sum of Definition 2.5 no larger than this one. So (20) is the genuine pushforward of a Borel probability measure by a Borel map into the carrier: $\mu_N(A) = \mu_N^{\mathrm{GJ}}(L_N^{-1}A)$ for Borel $A$, and $\mu_N$ is a Borel probability measure there for every $N$.

What (20) *is* is a carrier-valued representative of the lattice measure itself, and Lemma 5.8 records this as a lemma: $L_N$ is total on the configuration space, Borel, and lands in the carrier. What is *not* claimed anywhere is a statement about a total lift specified independently of $L_N$ — in particular the series-defined total lift of the machine development, where the measurability of that chosen function is an input of the data structure rather than a construction (Remark 5.9). The object of the law limit is $L_N$ and the measure $\mu_N$ of (20).

3.4. **Haar versus physical Lebesgue normalisation.**

**Lemma 3.7** (Haar-to-physical-Lebesgue normalisation)**.** *Let $f$ be a bounded continuous $L$-periodic test, put $f_L(z) := f(Lz)$ for $z \in [0,1)^2$, and let $\mathrm{d}z$ be Lebesgue measure on the unit cell, of total mass 1. The orthonormal coefficient of $f_L$ at $k \in \mathbb{Z}^2$ is*

$$(Bf_L)(k) := \int_{[0,1)^2} f(Lz)\, e^{-2\pi i\, k\cdot z}\, \mathrm{d}z = V^{-1}F(k), \tag{21}$$

*with $F$ the integral-normalised coefficient* (1) *of Definition 2.1.*

*Proof.* Substituting $x = Lz$, so that $\mathrm{d}x = L^2\mathrm{d}z$ and $k \cdot z = k \cdot x/L$, three steps give the claim:

$$\begin{aligned}(Bf_L)(k) &= \int_{(0,1]^2} f(Lz)\, e^{-2\pi i\, k\cdot z}\, \mathrm{d}z,\\ &= \int_{[0,L]^2} f(x)\, e^{-2\pi i\, k\cdot x/L}\, L^{-2}\, \mathrm{d}x,\\ &= V^{-1}\int_{[0,L]^2} f(x)\, e^{-2\pi i\, k\cdot x/L}\, \mathrm{d}x = V^{-1}F(k).\end{aligned}$$

The half-open cell may be replaced by the closed one because the two differ by a Lebesgue-null set; the phase identity $k \cdot (Lz) = L\,(k \cdot z)$ holds for negative $k$ as written, with no conjugation; and $L^2 = V$. □

This is the single easiest place in the paper to lose a factor of $L^d$, which in two dimensions is the volume $V$: a change of variables pays exactly one Jacobian, and substituting the orthonormal coefficient where the integral-normalised one is meant, inside a pairing that already carries its own $V^{-1}$ (Remark 2.2), loses that factor if it is done once and injects a spurious one if it is done twice.

3.5. **The two limits taken later.** Two limits are taken in what follows, and they are of different kinds. One is the covariance limit of Section 4: the lattice covariance pairings (18) of bounded continuous periodic tests are shown to converge as $N \to \infty$ at fixed $L$ and $m$, against an independently constructed resolvent. The other is the law limit of Section 6–Section 9: the object is the sequence $(\mu_N)$ of pushforwards (20), each a Borel probability measure on the carrier, and the question is what a limit of that sequence on that carrier is. Which topology carries it, and what a limit there means, is fixed in Section 5; the definitions above are what that section and the ones after it take as given.

## 4. The physical-volume covariance limit

This section proves the one analytic limit of Part I that is not measure theory: the covariance integrals of the physical lattice laws converge, and their limit is a pairing with a resolvent constructed before any Fourier series is written. Lemma 4.1 builds that resolvent from an energy form and a Riesz representation [16], Lemma 4.3 rewrites the lattice covariance of Definition 3.4 as a counting

pairing with the inverse of the Gaussian operator $A_N = m^2 - \Delta_a$, and Theorem 4.4 combines the two. The objects are the pairings

$$\mathcal{Q}_N(f,g) := \frac{1}{V}\sum_{p\in I_N} \frac{\overline{F_N(p)}\, G_N(p)}{\lambda_N(p)}, \qquad \mathcal{Q}(f,g) := \frac{1}{V}\sum_{k\in\mathbb{Z}^2} \frac{\overline{F(k)}\, G(k)}{\lambda(k)}, \tag{22}$$

where $F$ is (1), $F_N$ the sampled coefficient (14), and $\lambda_N, \lambda$ the symbols (7)–(8). The calligraphic letter keeps $Q$ for the Gaussian operator of Definition 3.1, and no other letter is free. Since $\overline{F_N(p)} = F_N(-p)$ and both symbols are even, the sums are real for real $f, g$; the computation is dimension-blind except in one shell count, flagged in Step 5 below, and we write the two-dimensional case that Section 2 fixes.

**Lemma 4.1** (The periodic massive resolvent)**.** *Let $\nu$ be Lebesgue measure on $[0,1)^2$, of total mass 1, and $E := L^2([0,1)^2, \nu; \mathbb{C})$ with $\langle u, v\rangle_E = \int \overline{u}\, v \,\mathrm{d}\nu$, conjugate-linear in the first argument. Write $b_k(z) = e^{2\pi i\, k\cdot z}$ for the basis of external theorem 2.7 and $B : E \to H$ for the induced isometry onto $H := \ell^2(\mathbb{Z}^2; \mathbb{C})$. Put $\lambda(k) = m^2 + \sum_i (2\pi k_i/L)^2$ and $w(k) := \lambda(k)^{-1/2}$, and let $J$ and the gradient multipliers $D_i$ act diagonally by $(Jh)(k) = w(k)h(k)$ and $(D_i h)(k) = i\,(2\pi k_i/L)\, w(k)h(k)$; the value and gradient maps are $jh := B^{-1}Jh$ and $B^{-1}D_i h$. Then:*

(i) *the sesquilinear form*

$$\mathcal{E}(h,v) := m^2 \langle jh, jv\rangle_E + \sum_{i=1}^{2} \langle D_i h, D_i v\rangle_E \tag{23}$$

*is the energy form of $H^1(T_L^2)$ and satisfies $\mathcal{E}(h,v) = \langle h, v\rangle_H$ for all $h, v \in H$, and the physical-volume inner product on the torus of period $L$ is $V\langle\cdot,\cdot\rangle_E$;*

(ii) *$\{(jh, (D_i h)_i) : h \in H\}$ is the closure in $E \times E^2$ of the pairs formed by finite trigonometric polynomials on $T_L^2$ and their classical physical-coordinate derivatives; this is the periodic $H^1(T_L^2)$ graph completion of the energy form in (i);*

(iii) *for $g \in E$ the vector $h_g := J^* Bg$, $Rg := jh_g$, is the unique element of $E$ satisfying*

$$m^2 \langle Rg, jv\rangle_E + \sum_{i=1}^{2} \langle D_i h_g, D_i v\rangle_E = \langle g, jv\rangle_E \tag{24}$$

*for every $v \in H$, and $\|Rg\|_E \le m^{-2}\|g\|_E$;*

(iv) *$B(Rg)(k) = \lambda(k)^{-1} Bg(k)$, and $\langle f, Rg\rangle_E = \sum_k \overline{Bf(k)}\, Bg(k)/\lambda(k)$, the series being absolutely convergent.*

*The construction is carried out on the unit-mass $L^2$ of the torus. No distribution-defined Sobolev space is invoked: the completion in (ii) is an explicit graph closure of polynomials.*

*Proof.* (i) The multipliers are diagonal in the basis $b_k$, and

$$m^2 w(k)^2 + \sum_i (2\pi k_i/L)^2 w(k)^2 = \lambda(k) w(k)^2 = 1$$

gives (23). The identity concerns the mixed sesquilinear form, not only its diagonal; physical volume enters only through $\mathrm{d}x = V\mathrm{d}\nu$. (ii) $J$ is bounded and injective because $0 < w(k) \le m^{-1}$, and finite-support vectors are dense in $H$. A finite-support $h$ gives the polynomial $\sum_k a(k) b_k$ with $a(k) = w(k)h(k)$, whose coordinate derivative is $\sum_k i\,(2\pi k_i/L)\, a(k) b_k$; conversely every finite-support polynomial arises this way, as $w(k) \neq 0$. If value-gradient pairs are Cauchy in $E \times E^2$, (23) makes the energy vectors Cauchy, completeness of $H$ supplies their limit, and continuity of $j$ and $D_i$ identifies the graph limit; the range is closed. (iii) Since $\|jv\|_E \le m^{-1}\|v\|_H$, the functional $v \mapsto \langle g, jv\rangle_E$ is bounded on $H$ and represented by $h_g = J^* Bg$, and $\langle h_g, v\rangle_H = \langle g, jv\rangle_E$ with (23) gives (24). Two solutions differ by $u$ with $\mathcal{E}(u,v) = 0$ for every $v$; testing $v = u$ gives

$u = 0$, and injectivity of $j$ fixes the $E$-value. Testing (24) with $v = h_g$ and taking real parts gives $m^2 \|Rg\|_E^2 \le \operatorname{Re}\langle g, Rg\rangle_E \le \|g\|_E \|Rg\|_E$, and we divide when $\|Rg\|_E \neq 0$. (iv) The adjoint identity on a single basis vector gives $(J^*c)(k) = w(k)c(k)$ because $w$ is real, so $B(Rg)(k) = w(k)^2 Bg(k) = \lambda(k)^{-1} Bg(k)$; Cauchy–Schwarz and $\lambda \ge m^2$ make the coefficient series absolutely convergent. □

**External theorem 4.2** (Physical-cell Riemann sums converge)**.** Let $g : [0, L]^2 \to \mathbb{C}$ be continuous, with sup-norm modulus of continuity $\omega_g$, and partition $[0, L]^2$ into the $N^2$ cubes of side $a = L/N$. The sum of the lower-corner sample times $a^2$ over all cubes differs from $\int_{[0,L]^2} g \,\mathrm{d}x$ by at most $V\omega_g(a)$, so these Riemann sums converge; half-open cells give the same value, the shared boundaries being Lebesgue null. This is the Henstock–Kurzweil box-integral theory of the standard library, quoted and not re-proved.

**Lemma 4.3** (Lattice covariance versus resolvent pairing)**.** *Let $b, c \in \mathbb{R}^{I_N}$ and let $f, g$ be real $L$-periodic. Then*

$$\langle C_N b, C_N c\rangle = a^{-2} \left\langle b, A_N^{-1} c\right\rangle_{\text{counting}}, \qquad a^2 \left\langle f, A_N^{-1} g\right\rangle_{\text{counting}} = \mathcal{Q}_N(f, g). \tag{25}$$

*Proof.* The first identity is (17) of Definition 3.4, the inverse-mass form in which the lattice covariance is used: $C_N$ is an input object (Remark 3.5), and the power $a^{-2}$ sits in the pairing, not in a rescaling of the field. For the second identity, use the character eigenbasis of Proposition 3.2: a real lattice function $u$ has $u(x) = V^{-1} \sum_p F_N(p)\chi_p(x)$ because $N^2a^2 = L^2 = V$, and $Q\chi_p = \lambda_N(p)\chi_p$ is (13), with $A_N = Q$ acting in the counting inner product. Substituting both expansions into the bilinear counting pairing and using the orthogonality $\sum_{x \in I_N} \chi_p(x)\chi_q(x) = N^2 \mathbf{1}_{p+q\equiv 0}$ together with $G_N(-p) = \overline{G_N(p)}$ for real $g$ gives

$$a^2 \left\langle f, A_N^{-1} g\right\rangle_{\text{counting}} = a^2 V^{-2} N^2 \sum_p F_N(p)\, \overline{G_N(p)}\, \lambda_N(p)^{-1} = \overline{\mathcal{Q}_N(f, g)} = \mathcal{Q}_N(f, g),$$

the last equality because $\mathcal{Q}_N$ is real for real $f, g$. The powers of $a$ are not cancelled by hand: each sampled vector carries $a^2$ against the $a^{-2}$ of the pairing, and $N^2a^2 = V$ turns the counting sum into the normalised pairing. Together with (18), this writes the covariance pairing in the symbol form the limit needs. □

The measure below is the physical lattice law $\mu_N^{\mathrm{GJ}}$ of Definition 3.4, not the finite-source push-forward $\mu_N$ whose laws converge in Section 6–Section 9: what is proved here is a statement about pairings, and it enters Section 7 as the pairwise covariance input, since every finite cylinder coordinate of the lift is a finite linear combination of lattice source coordinates (Definition 3.6); the entrywise identification is separate (Lemma 7.3).

**Theorem 4.4** (Fixed-volume covariance limit)**.** *Let $L > 0$ and $m > 0$, and let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be bounded continuous and $L$-periodic. Let $a = L/N$, let $\mu_N^{\mathrm{GJ}}$ be the physical lattice law of Definition 3.4, write $f_L \in E$ for $z \mapsto f(Lz)$ as in Lemma 3.7, and let $b, c \in \mathbb{R}^{I_N}$ be the sampled vectors $b_x = a^2 f(ax)$, $c_x = a^2 g(ax)$. Then, as $N \to \infty$ with $L$ and $m$ fixed,*

$$\begin{aligned} \langle C_N b, C_N c\rangle &\longrightarrow \operatorname{Re} V \left\langle f_L, Rg_L\right\rangle_E, \\ \int \omega(f)\,\omega(g)\, \mu_N^{\mathrm{GJ}}(\mathrm{d}\omega) &\longrightarrow \operatorname{Re} V \left\langle f_L, Rg_L\right\rangle_E, \\ \int \big(\omega(f) - \mathbb{E}\omega(f)\big)\big(\omega(g) - \mathbb{E}\omega(g)\big)\, \mu_N^{\mathrm{GJ}}(\mathrm{d}\omega) &\longrightarrow \operatorname{Re} V \left\langle f_L, Rg_L\right\rangle_E, \end{aligned} \tag{26}$$

*where $\omega(f)$ is the test average of Definition 3.4 and $\mathbb{E}$ is the expectation under $\mu_N^{\mathrm{GJ}}$. Both means vanish identically in $N$, so the second and third limits coincide.*

*Proof.* By Lemma 4.3 the first line of (26) is $\mathcal{Q}_N(f,g)$, and the other two are the mixed second moment and the centred covariance of the same Gaussian family. It suffices to prove $\mathcal{Q}_N(f,g) \to \mathcal{Q}(f,g)$ and to identify $\mathcal{Q}$ with the resolvent pairing, in five labelled steps.

**Step 1 (fixed-mode Riemann sums).** Let $k \in \mathbb{Z}^2$ and let $p_N(k) \in I_N$ be its residue class. At the sampled physical point $t = ax$ the phase is exactly $e^{-2\pi i\, k\cdot t/L} = e^{-2\pi i\, k\cdot x/N}$, for either representative of the residue class, and $t \mapsto f(t)e^{-2\pi i\, k\cdot t/L}$ is uniformly continuous on the compact cell with sup-norm modulus of continuity $\omega_k$. External theorem 4.2 applied to that function gives

$$|F_N(p_N(k)) - F(k)| \le V\omega_k(a), \qquad \omega_k(a) \longrightarrow 0. \tag{27}$$

Continuity on the compact cell is what produces $\omega_k(a) \to 0$; boundedness alone would not, since a bounded function may oscillate across every cube of every mesh.

**Step 2 (dispersion).** The limit $\sin t/t \to 1$ gives $4(N/L)^2 \sin^2(\pi k_i/N) \to (2\pi k_i/L)^2$, hence $\lambda_N(p_N(k)) \to \lambda(k)$. Both symbols are at least $m^2 > 0$, so their products and quotients converge; and the residue map is injective on $\{k : \|k\|_\infty \le K\}$ once $N > 2K$, so for fixed $K$ the sum over the cutoff converges term by term, real parts included.

**Step 3 (uniform denominators and tails).** Choose for each residue its centered representative in $\{-\lfloor N/2\rfloor, \dots, \lceil N/2\rceil - 1\}^2$, which represents the even-$N$ Nyquist frequency once. The chord bound $\sin t \ge 2t/\pi$ on $[0, \pi/2]$ gives, for centered $k$,

$$\lambda_N(k) \ge m^2 + \frac{16\,\|k\|_\infty^2}{L^2}, \qquad \text{so that} \qquad R_K := m^2 + \frac{16K^2}{L^2} \tag{28}$$

bounds $\lambda_N$ once $\|k\|_\infty > K$. Write $\mathcal{Q}_{N,K}$ and $\mathcal{Q}_K$ for the sums (22) restricted to $\|k\|_\infty \le K$. Cauchy–Schwarz with the finite Parseval identity (Lemma 3.3) — where boundedness of $f$ enters, and the only place where it does — gives

$$|\mathcal{Q}_N - \mathcal{Q}_{N,K}| \le \frac{1}{VR_K}\Big(\sum_p |F_N(p)|^2\Big)^{1/2}\Big(\sum_p |G_N(p)|^2\Big)^{1/2} \le \frac{V\,\|f\|_\infty\,\|g\|_\infty}{R_K} \tag{29}$$

uniformly in $N$ and $K \ge 0$; when no modes remain outside the cutoff the left side is zero. On the continuum side $\|k\|_\infty \ge K+1$ gives $\lambda(k) \ge D_K := m^2 + (2\pi/L)^2(K+1)^2$, and the same Cauchy–Schwarz estimate on the annuli between symmetric boxes gives

$$|\mathcal{Q} - \mathcal{Q}_K| \le \frac{V\,\|f\|_\infty\,\|g\|_\infty}{D_K} \longrightarrow 0. \tag{30}$$

Because $4\pi^2 > 16$ the constant $D_K$ is at least $R_K$, so (30) yields the tail with the constant of the discrete estimate; that inequality is what lets one cutoff serve both sides.

**Step 4 (eventual Bessel bound).** Fix the box $S_K = \{k : \|k\|_\infty \le K\}$. For $N > 2K$ reduction modulo $N$ is injective on $S_K$ and identifies $S_K$ with the discrete cutoff, so nonnegativity and Lemma 3.3 give

$$\sum_{k\in S_K} |F_N(p_N(k))|^2 \le V^2\,\|f\|_\infty^2 \qquad (N > 2K). \tag{31}$$

The restriction $N > 2K$ is not a convenience. The bound is *false* as a statement about all $N$: a small grid aliases distinct integer modes onto one residue, so the left side counts one coefficient several times and can exceed the right side without bound. Since every finite $S \subseteq \mathbb{Z}^2$ lies in some $S_K$, the fixed-mode limits (27) in (31) give $\sum_{k\in S} |F(k)|^2 \le V^2\,\|f\|_\infty^2$ for every finite $S$, and bounded finite sums of a nonnegative series give square summability of $k \mapsto F(k)$. No continuous Bessel theorem and no Fourier completeness is assumed: this is the Bessel *inequality*, and it is finite-dimensional.

**Step 5 (closed form, summability, envelope, assembly).** By (iii)–(iv) of Lemma 4.1 and the Haar-to-Lebesgue normalisation $Bf_L(k) = V^{-1}F(k)$ of Lemma 3.7,

$$V\,\langle f_L, Rg_L\rangle_E = V\sum_k \overline{V^{-1}F(k)}\,\frac{V^{-1}G(k)}{\lambda(k)} = V^{-1}\sum_k \frac{\overline{F(k)}\,G(k)}{\lambda(k)} = \mathcal{Q}(f,g). \tag{32}$$

So the Fourier Green pairing already carries the physical factor $L^{-2}$, and the identification of $\mathcal{Q}$ with the resolvent pairing is a computation, not a definition of $\mathcal{Q}$ as an operator. Absolute summability follows from Young's inequality and $\lambda \geq m^2$,

$$\left|\frac{\overline{F(k)}\,G(k)}{\lambda(k)}\right| \leq \frac{|F(k)|^2 + |G(k)|^2}{m^2}, \tag{33}$$

so the series in (22) converges, and exhaustion by symmetric boxes gives $\mathcal{Q}_K \to \mathcal{Q}$.

A modewise majorant independent of $N$ needs more than continuity. Fix $r = d+1$, which is $r = 3$ for the case $d = 2$ treated here, and $D_r(f) := \max_i \|\partial_i^r f\|_\infty$. The $r$-fold forward difference at spacing $a$ is bounded by $a^r D_r(f)$ and multiplies the sampled sum by $(\chi - 1)^r$; on a nonzero centered frequency with $\|k\|_\infty = q$ the chord bound gives $\left|e^{2\pi i k_i/N} - 1\right| \geq 4q/N$, and $1/\lambda_N, 1/\lambda \leq L^2/(16q^2)$ yields the common envelope

$$|F_N(k)| \leq V\Big(\frac{L}{4}\Big)^r D_r(f)\, q^{-r}, \qquad \frac{1}{V}\left|\frac{\overline{F_N(k)}\,G_N(k)}{\lambda_N(k)}\right| \leq \frac{VL^2}{16}\Big(\frac{L}{4}\Big)^{2r} D_r(f) D_r(g)\, q^{-2r-2}. \tag{34}$$

The sup-norm shell $\|k\|_\infty = q$ carries $(2q+1)^d - (2q-1)^d$ modes, at most $2d(2q+1)^{d-1}$; in $d = 2$ the count is $8q$ (Lemma 2.8). Since $2r + 2 > d$, the shell majorant is summable, and the zero mode is bounded by $m^{-2}$ and $V\,\|f\|_\infty \|g\|_\infty$. The envelope (34) is a statement about $C^r$ periodic tests. It is *not* available for all continuous tests, and no such statement is claimed: the limit below uses no envelope, which is why Theorem 4.4 concerns bounded continuous periodic tests and is not a modewise convergence theorem.

Choose $K$ so that both tails (29) and (30) are below $\epsilon/3$; Step 2 then bounds the fixed-cutoff error by $\epsilon/3$ for all large $N$. The triangle inequality gives

$$|\mathrm{Re}\,\mathcal{Q}_N(f,g) - \mathrm{Re}\,\mathcal{Q}(f,g)| \leq |\mathrm{Re}\,\mathcal{Q}_N - \mathrm{Re}\,\mathcal{Q}_{N,K}| + |\mathrm{Re}\,\mathcal{Q}_{N,K} - \mathrm{Re}\,\mathcal{Q}_K| + |\mathrm{Re}\,\mathcal{Q}_K - \mathrm{Re}\,\mathcal{Q}| < \epsilon, \tag{35}$$

with one $K$ serving all $N$, and with no caller supplying summability, a tail estimate or a convergence hypothesis. Transferring (35) to the measures uses only that $\mu_N^{\mathrm{GJ}}$ is Gaussian with covariance $C_N$ (Definition 3.4) and has mean zero exactly at every mesh: then the mixed second moment and the centred covariance integral both equal $\mathrm{Re}\,\mathcal{Q}_N$ and hence converge to $\mathrm{Re}\,V\,\langle f_L, Rg_L\rangle_E$ by (32). This proves (26). □

*Remark* 4.5 (Why no Bessel kernel appears). The proof of Theorem 4.4 introduces no lattice Green function $G_a(x-y)$, no Bessel function, no modified Bessel function $K_0$, no constant $1/(2\pi)$ or $1/(4\pi)$, and no logarithmic divergence. Every estimate is an estimate on symbols: $\lambda_N^{-1}$ on $I_N$ and $\lambda^{-1}$ on $\mathbb{Z}^2$, compared in (28)–(30) and controlled by (31) and (33). Nothing is lost by staying in the symbol domain: what is proved is a convergence statement about pairings of bounded continuous periodic tests with the resolvent, and such a pairing is determined by the spectral data of the resolvent on the Fourier basis, which is exactly $\lambda^{-1}$; a positional kernel is that same data written in another coordinate system, and producing it in closed form is a special-function computation the argument neither performs nor needs. A text that wrote the two-dimensional kernel $C_m(x) = K_0(m\,|x|)/(2\pi)$ here would be naming the $x$-space representative of the *same* operator on the same torus — the two are equal by the classical Green-function identity, and evaluating (22) against that kernel reproduces $\mathrm{Re}\,Q$ — but that identity is neither used above nor proved above, and this paper does not prove it; what is proved is the limit of the pairings, in the symbol domain, with the constants of

this section. Nor is a logarithmic divergence at issue: at fixed volume $\lambda_N(0) = \lambda(0) = m^2$, and the high-frequency behaviour is absorbed by (33) through the test coefficients rather than by a kernel.

With Theorem 4.4 the analysis of the lattice symbol is finished. Every later step of Part I is measure theory on the carrier of Section 5: the uniform moment and compact tail of Section 6, the finite marginals of Section 7, and the assembly in Section 9. No further symbol estimate occurs, and none is needed.

## 5. The carrier and its Borel structure

This section builds the space on which the lattice laws of Section 3 are Borel probability measures, so that convergence below is asserted against bounded continuous functions on a concrete space. That space is an explicit weighted sequence space over the mode lattice, and the properties a limit theorem normally assumes of its ambient space are proved for it here: completeness and separability (Definition 5.1 and Theorem 5.6), continuity and point separation of the coordinate maps (Lemma 5.5), and the equality of the sigma algebra generated by finitely many coordinates with the Borel sigma algebra (Theorem 5.7). Proposition 5.4 records the comparison of the two weights that makes the compactness argument of Section 6 possible.

No moment is estimated and no tightness is proved here; that is Section 6. Nor is Lemma 5.8 proved here: it is stated below and proved from facts of Section 3 and of this section.

### 5.1. **The carrier.**

**Definition 5.1** (The carrier)**.** The *carrier* is the weighted sequence space

$$\mathcal{C} := H^{-3} = \Big\{ c = (c_k)_{k\in\mathbb{Z}^2} \in \mathbb{C}^{\mathbb{Z}^2} : \sum_{k\in\mathbb{Z}^2} w_3(k)\,|c_k|^2 < \infty \Big\}, \qquad \|c\|_{\mathcal{C}}^2 := \sum_{k\in\mathbb{Z}^2} w_3(k)\,|c_k|^2\,, \tag{36}$$

with $w_3$ the weight (2) of Definition 2.3, that is $w_3(0) = 1$ and $w_3(k) = \|k\|_\infty^{-6}$ for $k \neq 0$. The space $H^{-3}$ of Definition 2.5 is spelled $\mathcal{C}$ from here on; it is the space the measures below live on, and $\|\cdot\|_{\mathcal{C}}$ is the norm in which they are tight. Writing $\widehat{c} := (\sqrt{w_3(k)}\,c_k)_{k\in\mathbb{Z}^2}$ for the *weighted family* of $c$, the map $\Theta : \mathcal{C} \to \ell^2(\mathbb{Z}^2)$, $\Theta c = \widehat{c}$, is a linear isometry of $\mathcal{C}$ onto the unweighted space of complex square-summable families on the mode lattice: the carrier is $\ell^2$ of the mode lattice in weighted coordinates. The weighted family is what the carrier stores — the completion of Definition 2.5 is realised as an $\ell^2$ space whose coordinate at $k$ is $\sqrt{w_3(k)}\,c_k$ — so below the *coordinates* of a carrier point are the entries of $\widehat{c}$, while $c$ is the family of Fourier coefficients. Its index set is $\mathbb{Z}^2$ throughout: the carrier, its weight and (Remark 2.9) the measure chain are two-dimensional, and no general-dimensional carrier is defined here.

*Remark* 5.2 (Which family a formula is written in). Two families denote the same carrier point and the paper uses both, so the rule is stated once. The *coefficient* family $c$ is what the symbols and pairings of Section 4 are written for, and its norm is (36). The *stored* family $\widehat{c} = \Theta c$ is what a formula written in coordinates uses — the Fourier lift of Definition 3.6, the moment of Section 6, and the cylinder maps of Section 7 — and every weight those formulae carry is the one transported along $\Theta$, so their coordinates already include it. Because $\Theta$ is an isometry, a computation may be done in either family; the only error available is to apply the weight twice, and no formula below does: a factor $\sqrt{w_3(k)}$ appears in a coordinate formula exactly when the coefficient at $k$ is being turned into a coordinate, and nowhere else.

*Remark* 5.3 (One spelling for three source notations). The sources write this one space in three ways: as `lp` ((`Fin 2` $\to \mathbb{Z}$) $\to \mathbb{C}$) 2, in its curried form `lp` (`Fin 2` $\to \mathbb{Z} \to \mathbb{C}$) 2, and as the same `lp` space over a constant family of fibres, the fibres being indexed by the type that the sources call `PeriodicFourierIndex`. The second is the curried form of the first, the third differs only in presenting the family of fibres as a function of the index, and all three denote the same weighted

$\ell^2$ space; the machine development names it `PeriodicHMinusThreeCompletion`. This paper writes $\mathcal{C}$ and nothing else.

**Proposition 5.4** (Completion and embedding)**.**

(i) *$\mathcal{C}$ is complete. Equivalently, the completion of the finitely supported families for the norm $\|\cdot\|_{\mathcal{C}}$ is $\mathcal{C}$ itself, and $\Theta$ identifies it isometrically with $\ell^2(\mathbb{Z}^2)$.*

(ii) *The weights are pointwise ordered, $w_3(k) \le \widetilde{w}_2(k)$ for every $k \in \mathbb{Z}^2$, and consequently*

$$\|c\|_{\mathcal{C}} \le \|c\|_{H^{-2}} \qquad \text{for every } c \in H^{-2}, \tag{37}$$

*so $H^{-2} \subseteq \mathcal{C}$ and the inclusion is continuous with operator norm one, attained at every mode with $\|k\|_\infty = 1$. Moreover $w_3(k)/\widetilde{w}_2(k) = \|k\|_\infty^{-2}$ tends to 0 along the mode lattice.*

(iii) *Let $S \subseteq H^{-2}$ with $\sup_{c\in S} \|c\|_{H^{-2}} \le B < \infty$. Then for every integer $R \ge 0$ the members of $S$ have uniformly small tails in the carrier norm:*

$$\sup_{c\in S} \sum_{\|k\|_\infty > R} w_3(k)\,|c_k|^2 \;\le\; \frac{B^2}{(R+1)^2}\ . \tag{38}$$

*The compactness statement corresponding to* (38)*, that the closure of such a set is compact in $\mathcal{C}$, is Theorem 6.4.*

*Proof.* (i) $\Theta$ is a linear isometry of $\mathcal{C}$ onto the complete space $\ell^2(\mathbb{Z}^2)$, and a linear isometry onto a complete space has complete domain, since Cauchy sequences and limits transport along an isometry.

(ii) For $k \neq 0$ we have $\|k\|_\infty \ge 1$, hence $\|k\|_\infty^{-6} \le \|k\|_\infty^{-4}$; at $k = 0$ both weights equal 1. Summing against $|c_k|^2$ gives (37). Since $\|e_k\|_{\mathcal{C}} = \|e_k\|_{H^{-2}} = 1$ at every mode of $\ell^\infty$-norm one, the operator norm is exactly one.

(iii) For integer $R$ and $\|k\|_\infty > R$ we have $\|k\|_\infty \ge R+1$, so $w_3(k) \le (R+1)^{-2}\widetilde{w}_2(k)$ there. Hence

$$\sum_{\|k\|_\infty > R} w_3(k)\,|c_k|^2 \le \frac{1}{(R+1)^2} \sum_{k\in\mathbb{Z}^2} \widetilde{w}_2(k)\,|c_k|^2 = \frac{\|c\|_{H^{-2}}^2}{(R+1)^2} \le \frac{B^2}{(R+1)^2},$$

the middle identity being the definition of the $H^{-2}$ norm, which uses the full weight and so includes the zero mode. □

The pair $(2, 3)$ is fixed by two different requirements in dimension two. Gaussian second-moment summability in $H^{-s}$ needs $s > (d-2)/2$, which in $d = 2$ asks only for $s > 0$, whereas relative compactness of the corresponding sublevel sets needs a strictly weaker ambient exponent $r > s$; the pair $(s, r) = (2, 3)$ satisfies both with room to spare. Only this two-dimensional instance is used below.

## 5.2. **Coordinates, separability, and cylinders.**

**Lemma 5.5** (Coordinate embeddings)**.**

(i) *For every $k \in \mathbb{Z}^2$ the coordinate functional $\pi_k : \mathcal{C} \to \mathbb{C}$, $\pi_k(c) = c_k$, is continuous, with $|\pi_k(c)| \le w_3(k)^{-1/2}\,\|c\|_{\mathcal{C}}$.*

(ii) *For every $m \ge 1$, every $m$-tuple of modes and every choice of a real or an imaginary part at each slot, the map $\mathcal{C} \to \mathbb{R}^m$ whose components are those parts of $c_{k_1}, \dots, c_{k_m}$ is continuous, and so is the complex version $\mathcal{C} \to \mathbb{C}^m$, $c \mapsto (c_{k_1}, \dots, c_{k_m})$.*

(iii) *The coordinates separate points: if $c \neq c'$ in $\mathcal{C}$ then $c_k \neq c'_k$ for some $k \in \mathbb{Z}^2$.*

*Proof.* (i) $w_3(k)\,|c_k|^2$ is a single term of the sum (36), and $w_3(k) > 0$ at every mode. (ii) Each component composes the continuous functional $\pi_{k_i}$ with one of the continuous maps $z \mapsto \operatorname{Re} z$, $z \mapsto \Im z$, and a map into a finite product is continuous exactly when its components are. (iii) If $c_k = c'_k$ for every $k$ the two families coincide. □

Point separation is weaker than determination of the Borel structure, and the difference matters. Continuity of the coordinates makes the sigma algebra they generate a sub-sigma-algebra of the Borel sigma algebra, and separation of points makes the coordinate map injective, but neither says that this sub-sigma-algebra is the whole Borel sigma algebra: for a metric space that is not separable the conclusion can fail. The next two results supply separability first and the sigma algebra equality second.

**Theorem 5.6** (Separability)**.** *The carrier contains a countable dense subset. Consequently it is a complete separable metric space, hence Polish, so Prokhorov's theorem applies to families of Borel probability measures on it (external theorem 6.6).*

*Proof.* Fix an enumeration $k_1, k_2, \dots$ of $\mathbb{Z}^2$, and for $m \geq 1$ let $T_m c := \sum_{j \leq m} c_{k_j} e_{k_j}$ keep the first $m$ coordinates in that enumeration and set the rest to zero. Let $D$ be the set of families $\sum_{j \leq m} q_j e_{k_j}$ with $m \geq 1$ and $q_j \in \mathbb{Q} + i\mathbb{Q}$; as a countable union of countable sets, $D$ is countable.

*The truncations converge in the carrier norm.* Let $c \in \mathcal{C}$. By (36) the series $\sum_{j \geq 1} w_3(k_j) \left|c_{k_j}\right|^2$ is summable, hence converges as a series of non-negative terms, and therefore

$$\|c - T_m c\|_{\mathcal{C}}^2 = \sum_{j > m} w_3(k_j) \left|c_{k_j}\right|^2 \longrightarrow 0 \qquad (m \to \infty). \tag{39}$$

This is where summability enters, and it is the only place: $c - T_m c$ is the tail of that same series, its squared carrier norm is exactly that tail, and the tail of a convergent series of non-negative terms tends to zero. What is summed is therefore the weight against one square-summable family, and (39) is convergence in the carrier norm.

*Rational approximation at fixed truncation.* Fix $m$ and $\varepsilon > 0$. Since the sum in (39) is over the finite set $k_1, \dots, k_m$, choose $q_1, \dots, q_m \in \mathbb{Q} + i\mathbb{Q}$ with $w_3(k_j) \left|c_{k_j} - q_j\right|^2 \leq \varepsilon^2/m$ for $j \leq m$; then $d := \sum_{j \leq m} q_j e_{k_j} \in D$ satisfies $\|T_m c - d\|_{\mathcal{C}} \leq \varepsilon$.

Given $c$ and $\varepsilon > 0$, choose $m$ with $\|c - T_m c\|_{\mathcal{C}} < \varepsilon$ and then $d \in D$ with $\|T_m c - d\|_{\mathcal{C}} < \varepsilon$, so $\|c - d\|_{\mathcal{C}} < 2\varepsilon$. Hence $D$ is dense and $\mathcal{C}$ is separable. Completeness is Proposition 5.4(i), and a complete separable metric space is Polish. The machine development proves this statement as `periodicHMinusThreeCompletion_separable`, one of the results listed in Appendix A. □

**Theorem 5.7** (Cylinder sigma algebra equals Borel)**.** *Let $\Sigma_{\mathrm{cyl}}$ be the sigma algebra on $\mathcal{C}$ generated by the coordinate functionals $\pi_k$, $k \in \mathbb{Z}^2$: the smallest sigma algebra containing $\Pi^{-1}(B)$ for every finite coordinate map $\Pi$ of Lemma 5.5 and every Borel $B \subseteq \mathbb{C}^m$. Then $\Sigma_{\mathrm{cyl}} = \mathcal{B}(\mathcal{C})$.*

*Proof.* Write $E : \mathcal{C} \to \mathbb{C}^{\mathbb{Z}^2}$, $E(c) = (c_k)_{k \in \mathbb{Z}^2}$, for the coordinate map, and let $\mathcal{B}_\Pi$ be the product sigma algebra on $\mathbb{C}^{\mathbb{Z}^2}$. Since $\mathbb{Z}^2$ is countable, $\mathcal{B}_\Pi$ is generated by the coordinate projections, so

$$\Sigma_{\mathrm{cyl}} = E^{-1}(\mathcal{B}_\Pi) \tag{40}$$

by definition.

*The cylinders are Borel.* Each $\pi_k$ is continuous by Lemma 5.5(i), hence Borel, so $\pi_k^{-1}(B)$ is Borel for Borel $B$; the sets with Borel preimage under $E$ form a sigma algebra, and $\Sigma_{\mathrm{cyl}} \subseteq \mathcal{B}(\mathcal{C})$ follows from (40).

*The Borel sets are cylindrical.* By Theorem 5.6 the carrier is separable, so its topology has a countable base: with a countable dense set $D_0$ fixed, the balls $B(q, r)$, $q \in D_0$, $r \in \mathbb{Q}$, $r > 0$, form one. A sigma algebra containing every base element contains every open set, hence every Borel set. So it suffices to place every ball in $\Sigma_{\mathrm{cyl}}$. Fix $c_0 \in \mathcal{C}$, $r > 0$ and put $s_m(c) := \sum_{j \leq m} w_3(k_j) \left|c_{k_j} - c_{0,k_j}\right|^2$. Each $s_m$ is a finite sum of functions of one coordinate, hence is $\Sigma_{\mathrm{cyl}}$-measurable, and $s_m(c) \uparrow \|c - c_0\|_{\mathcal{C}}^2$ because the series (36) converges. Therefore

$$\overline{B(c_0, r)} = \bigcap_{n \geq 1} \bigcup_{M \geq 1} \bigcap_{m \geq M} \left\{ c \in \mathcal{C} : \; s_m(c) \leq (r + 1/n)^2 \right\}$$

is a countable Boolean combination of $\Sigma_{\mathrm{cyl}}$-measurable sets, and the equality of sets follows from that monotone convergence: a point of the closed ball satisfies every bound, while a point outside it violates the bound at any $n$ with $r + 1/n < \|c - c_0\|_{\mathcal{C}}$, once $m$ is large. Finally $B(c_0, r) = \bigcup_n \overline{B(c_0, r - 1/n)}$ over those $n$ with $r - 1/n > 0$, so every open ball, hence every open set, lies in $\Sigma_{\mathrm{cyl}}$. □

The cylinders themselves form a $\pi$-system, the intersection of two of them being the cylinder over the combined tuple, and because $\mathbb{Z}^2$ is countable that family generates $\Sigma_{\mathrm{cyl}}$; Theorem 5.7 therefore exhibits a $\pi$-system generating the Borel sigma algebra, which is what the determination theorem of Section 8 consumes through the extension argument of Dynkin.

5.3. **The lift, and the topology of convergence.** Section 3 already exhibits a total carrier-valued representative of the physical lattice law; this section states it once as a lemma, so that no later section has to hedge about it.

**Lemma 5.8** (The finite-source lift is a carrier-valued representative)**.** *Let $N \geq 1$ and let $\mu_N^{\mathrm{GJ}}$ be the physical lattice Gaussian law of Definition 3.4. Then the finite-source Fourier lift $\mathcal{L}_N : \mathbb{R}^{I_N} \to \mathbb{C}^{\mathbb{Z}^2}$ of Definition 3.6 takes values in $\mathcal{C}$ for* every *configuration, it is Borel as a map into $\mathcal{C}$, and its entries are*

$$\big(\mathcal{L}_N(\omega)\big)_k = \sqrt{w_3(k)}\, a^2 \sum_{x \in I_N} \omega(x)\, e^{-2\pi i\, k\cdot(ax)/L} \qquad (\omega \in \mathbb{R}^{I_N},\ k \in \mathbb{Z}^2). \tag{41}$$

*Consequently $\mu_N = (\mathcal{L}_N)_* \mu_N^{\mathrm{GJ}}$ is a Borel probability measure on $\mathcal{C}$ that carries the law of the lattice field, and no exceptional set is involved.*

*Proof.* Three facts, each established before this point. *Range and finiteness:* by (19), $\mathcal{L}_N\omega$ is a finite real-linear combination of the coordinate evaluations $\omega \mapsto \omega(x)$, and $\|\mathcal{L}_N\omega\|_{\mathcal{C}}^2$ is finite for every $\omega$ by the weight sum computed in Section 3, where it is bounded by $a^4 \|\omega\|_1^2 \big(1 + \sum_{p \neq 0} \|p\|_\infty^{-6}\big)$. *Measurability:* each coordinate $c \mapsto c_k$ is continuous (Lemma 5.5), hence Borel, and those coordinates generate $\mathcal{B}(\mathcal{C})$ (Theorem 5.7); a map into a separable metric space whose coordinates are Borel is Borel. *The entry formula* is (19), which carries the factor $\sqrt{w_3}$ of the stored representation; the entries are therefore the stored ones, not the raw coefficients, and Remark 5.2 is the dictionary between the two. The last sentence is the definition of $\mu_N$ in Eq. (20) together with those three facts. □

*Remark* 5.9 (What the formalisation does with the lift). The development of Section 11 does not use $\mathcal{L}_N$ for this purpose. Its carrier-valued lift is a *total* function, defined coordinatewise (the series appears only in the $\ell^2$ side condition that the completion imposes, not in the definition), together with the convention that the value is 0 off the convergence set. Its measurability is carried as one conjunct of an input of the formal development rather than derived there; the other conjunct asserts that this chosen function agrees with the raw weighted lift almost everywhere, so that input does speak to the size of the exceptional set. Three things follow, and all are about the formalisation. First, the input is a statement about one chosen total function, not about the existence of a carrier-valued representative, which Lemma 5.8 settles outright. Second, what the input supplies is a.e.-measurability and a.e.-agreement, whereas the lemma supplies a range statement for every $\omega$ and genuine Borel measurability; a reproducer reading only the input should not read it as the lemma in other notation. Third, nothing here turns on that difference on the finite-source route: the development itself proves, with no input, that the composition of the total lift with the lattice embedding is measurable, and that its pushforward coincides with the finite-source pushforward. A reader who compares the two should read Theorem 9.1 as a statement about $\mu_N$, which is what it is.

*Remark* 5.10 (The topology used for convergence). Convergence below is weak convergence of Borel probability measures on the carrier [1]: $\mu_n \Rightarrow \mu$ means $\int \varphi \, \mathrm{d}\mu_n \to \int \varphi \, \mathrm{d}\mu$ for every bounded continuous $\varphi : \mathcal{C} \to \mathbb{R}$. This is the topology Theorem 5.6 makes available. Three things are deliberately avoided: a weak-star topology on a dual, since the measures are Borel laws on $\mathcal{C}$ itself and $\mathcal{C}$ is never treated as the dual of a test space; a metric on the space of probability measures, although a Prokhorov metric exists; and a topology on a space of test functions, since what is paired with the measures is a bounded continuous function on the carrier, never a distribution. The operational criterion used later is not that convergence of finitely many coordinate marginals implies weak convergence, which is false without a tightness hypothesis, but that a weak limit, whose existence comes from tightness together with external theorem 6.6, is determined by its finite coordinate marginals: continuity of the cylinder maps transfers the marginal limits to any limit point, and Theorem 5.7 supplies the sigma algebra equality the determination argument needs. Tightness itself is the content of Section 6, not of this section.

## 6. Uniform moment bound and tightness

This section supplies the compactness input that Prokhorov's theorem consumes, and it derives the uniform trace constant rather than quoting it. The constant is Lemma 6.1, the moment it feeds is Theorem 6.2, the compactness of the sets that moment controls is Theorem 6.4, and Markov's inequality combines the two into tightness in Theorem 6.5; Lemma 6.3 separates the two weight conventions the argument needs. Nothing here concerns the marginals of $\mu_N$ or a limit.

Fix $L > 0$, $m > 0$ and $N \in \mathbb{N}$, $a > 0$ with $aN = L$ (Hypothesis 2.6); along the sequence $N = n+1$ and $a = L/N$. Let $\cos_k(x) = \cos(2\pi\, k \cdot x/L)$ and $\sin_k(x) = \sin(2\pi\, k \cdot x/L)$ be the cosine and sine tests of mode $k \in \mathbb{Z}^2$, and extend the test pairing $\varphi(f)$ of a lattice field, which is real for real $f$, to complex tests by complex linearity. The *probe* of the field at $k$ is

$$\widehat{\varphi}(k) := \varphi(\cos_k) - i\,\varphi(\sin_k) \in \mathbb{C}, \tag{42}$$

that is, the field paired with the complex exponential of mode $k$ in the normalisation (1). Only $|\widehat{\varphi}(k)|^2 = \varphi(\cos_k)^2 + \varphi(\sin_k)^2$ enters below. The hat records the mode; it is not the weighted family $\widehat{c}$ of Definition 5.1, which records the weight. At $k = 0$, $\cos_0 \equiv 1$ and $\sin_0 \equiv 0$, so $\widehat{\varphi}(0) = \varphi(1)$. The stored coordinate of a carrier point at $k$ is $\sqrt{w_3(k)}$ times the probe (Eq. (19)), while the $H^{-2}$ norm below is written in the coefficient family of Remark 5.2 and carries the $\widetilde{w}_2$ weight itself, so the two weights are never both applied; it is the $\widetilde{w}_2$-weighted sum of squared probe moduli,

$$\|\varphi\|^2_{H^{-2}} = \sum_{k \in \mathbb{Z}^2} \widetilde{w}_2(k)\, |\widehat{\varphi}(k)|^2 \in [0, \infty], \tag{43}$$

finite on the $H^{-2}$ subspace and $+\infty$ outside it: the norm of Definition 2.5 with the full weight.

The lattice covariance is an input object here, and the only property of it that this section uses is its sampled covariance form Eq. (18). For bounded continuous tests $f, g$ it reads $\mathbb{E}_{\mu_N}[\varphi(f)\varphi(g)] = a^2 \left\langle f, A_N^{-1} g \right\rangle_{\text{counting}}$, in the counting inner product (12) and with $A_N = Q$ the operator of Definition 3.1. Its symbol is at least $m^2$ (Proposition 3.2), so $A_N \geq m^2$, and for every such $f$

$$\mathbb{E}_{\mu_N}\left[\varphi(f)^2\right] \leq a^2 m^{-2} \sum_{x \in I_N} f(ax)^2 \leq m^{-2}(aN)^2\, \|f\|^2_\infty = m^{-2} L^2\, \|f\|^2_\infty\,, \tag{44}$$

the last identity being (5).

### 6.1. The uniform trace constant.

**Lemma 6.1** (The uniform $H^{-2}$ trace constant). *Let $L > 0$, $m > 0$ and*

$$C(L, m) := m^{-2} L^2 + 2\, m^{-2} L^2 \sum_{k \in \mathbb{Z}^2} w_2(k). \tag{45}$$

*Then $C(L,m)$ is a finite real number with $m^{-2}L^2 \le C(L,m) \le 25\,m^{-2}L^2$, and for every $N$, $a$ with $aN = L$,*

$$\sum_{k\in\mathbb{Z}^2} \widetilde{w}_2(k)\,\mathbb{E}_{\mu_N}\big[|\widehat{\varphi}(k)|^2\big] \le C(L,m). \tag{46}$$

*The summand of (46) at $k = 0$ equals $m^{-2}L^2$ exactly; each summand with $k \neq 0$ is at most $2\,m^{-2}L^2 w_2(k)$.*

*Proof. (i) The per-mode second moment.* For $k \neq 0$ the tests $\cos_k$ and $\sin_k$ are bounded by 1 in sup norm, so (44) gives $\mathbb{E}[\varphi(\cos_k)^2] \le m^{-2}L^2$ and $\mathbb{E}[\varphi(\sin_k)^2] \le m^{-2}L^2$; by (42) and linearity in each real coordinate,

$$\mathbb{E}\big[|\widehat{\varphi}(k)|^2\big] = \mathbb{E}\big[\varphi(\cos_k)^2\big] + \mathbb{E}\big[\varphi(\sin_k)^2\big] \le 2\,m^{-2}L^2 = 2Vm^{-2}.$$

The factor 2 is exactly the pair of real coordinates: the squared modulus of a complex random variable is the sum of the squares of its real and imaginary parts, and each part is a real Gaussian image of the field with variance at most $Vm^{-2}$, because $\lambda_N \ge m^2$.

*(ii) The zero mode, exactly.* At $k = 0$ the sine test vanishes and the cosine test is the constant 1, so $\mathbb{E}[|\widehat{\varphi}(0)|^2] = \mathbb{E}[\varphi(1)^2] = a^2\,\langle 1, A_N^{-1}1\rangle_{\text{counting}}$ by (18). Constants are annihilated by the discrete Laplacian (Definition 3.1), so $A_N 1 = m^2 1$ and $A_N^{-1}1 = m^{-2}1$, while the counting inner product of 1 with itself is the number of sites $N^2$. Hence

$$\mathbb{E}\big[|\widehat{\varphi}(0)|^2\big] = a^2 m^{-2} N^2 = m^{-2}(aN)^2 = m^{-2}L^2 = V/m^2,$$

an equality and not an estimate: it is the covariance pairing of the constant test with itself. Only the cosine coordinate survives at $k = 0$, which is why this term carries no factor 2.

*(iii) Summability, with a value.* The shell $\{\|k\|_\infty = q\}$ carries $8q$ modes, on which $w_2 = q^{-4}$ (Lemma 2.8), hence

$$\sum_{k\in\mathbb{Z}^2} w_2(k) = \sum_{q\ge 1} 8q \cdot q^{-4} = 8\sum_{q\ge 1} q^{-3} \le 8\Big(1 + \int_1^\infty t^{-3}\,\mathrm{d}t\Big) = 12,$$

the summand at $k = 0$ contributing nothing. So the series in (45) converges, and only positivity of $m$ and $L$ enters this step.

*(iv) Assembly.* Since $\widetilde{w}_2(0) = 1$ and $\widetilde{w}_2(k) = w_2(k)$ for $k \neq 0$ (Definition 2.3) and probe moments are nonnegative, splitting at the zero mode and using (ii) on the first piece and (i) termwise on the second,

$$\sum_{k\in\mathbb{Z}^2} \widetilde{w}_2(k)\mathbb{E}\big[|\widehat{\varphi}(k)|^2\big] = \mathbb{E}\big[|\widehat{\varphi}(0)|^2\big] + \sum_{k\neq 0} w_2(k)\mathbb{E}\big[|\widehat{\varphi}(k)|^2\big] \le m^{-2}L^2 + 2\,m^{-2}L^2 \sum_{k\neq 0} w_2(k) = C(L,m),$$

the last sum being the one in (45) because $w_2(0) = 0$. The series on the left converges because its majorant does, which is (iii). □

**Theorem 6.2** (Uniform $H^{-2}$ moment)**.** *Let $L > 0$, $m > 0$ and let $C(L,m)$ be the constant (45). For every $N$, $a$ with $aN = L$, with $\mu_N$ the law (20) of the finite-source lift,*

$$\mathbb{E}_{\mu_N}\,\|\varphi_N\|_{H^{-2}}^2 \le C(L,m), \qquad \textit{hence} \qquad \sup_N \mathbb{E}_{\mu_N}\,\|\varphi_N\|_{H^{-2}}^2 \le C(L,m).$$

*The constant is independent of $N$: it is a function of $L$ and $m$ alone.*

*Proof.* By (43) the left side is the expectation of a countable sum of nonnegative continuous functions of the carrier point, a probe being the stored coordinate divided by the fixed positive number $\sqrt{w_3(k)}$ (Eq. (19)). Tonelli's theorem interchanges sum and expectation, and (46) bounds the result by $C(L,m)$. For the uniformity, $aN = L$ holds at every index by Hypothesis 2.6 and (45) contains neither $N$ nor $a$ nor $n$, so the bound is the same at every index and the supremum leaves it unchanged. □

6.2. **The zero mode.**

**Lemma 6.3** (The zero mode)**.** *Let $Z := \{\varphi : \widehat{\varphi}(k) = 0$ for every $k \neq 0\}$ be the zero-mode line and $p(\varphi) := \sum_k w_2(k)\,|\widehat{\varphi}(k)|^2$.* (i) *For every carrier point $\varphi$,*

$$\sum_{k\in\mathbb{Z}^2} \widetilde{w}_2(k)\,|\widehat{\varphi}(k)|^2 = |\widehat{\varphi}(0)|^2 + \sum_{k\in\mathbb{Z}^2} w_2(k)\,|\widehat{\varphi}(k)|^2 .$$

(ii) *$p$ annihilates $Z$ and is not a norm; for every $R \geq 0$ the set $\{p \leq R\}$ contains $Z$, whose closure in the carrier is not compact.* (iii) *The energy* (43) *restricted to $Z$ equals $|\widehat{\varphi}(0)|^2$.*

*Proof.* (i) is $\widetilde{w}_2 = w_2 + \mathbf{1}_{\{0\}}$ read termwise, the two cases being $w_2(0) = 0$ and $\widetilde{w}_2(0) = 1$. For (ii): if $\varphi \in Z$ every summand of $p$ vanishes, so $p$ has non-trivial kernel and is only a seminorm; $Z$ is a closed unbounded line contained in $\{p \leq R\}$, and no closed subset of the carrier containing an unbounded line is compact. For (iii): on $Z$ only the $k = 0$ summand survives, with weight $\widetilde{w}_2(0) = 1$. □

The lemma forces the two usages. The energy that defines the sets of Theorem 6.4 must be (43), with the full weight: by (ii) a sublevel set of the seminorm $p$ contains the unbounded line $Z$, so its closure cannot be compact, and tightness built on $p$ would have no compact set to offer. The zero mode is therefore paid for by the energy, and the value $\widetilde{w}_2(0) = 1$ is that payment.

The trace in Lemma 6.1 is the opposite case: the seminorm weight is used there because the zero mode is settled separately and exactly. By (i) the full weighted trace splits into the zero-mode moment plus a $w_2$-weighted remainder, and by step (ii) of Lemma 6.1 that moment equals $m^{-2}L^2$ rather than merely being bounded by $2Vm^{-2}$. Writing the constant with $w_2$ and adding $m^{-2}L^2$ back is thus the exact decomposition of the full trace, with the zero mode counted once and without a factor 2; applying the uniform per-mode bound at $k = 0$ as well would have lost the sharp value $V/m^2$. So the zero mode is included in the energy, which is what makes the sets of Theorem 6.4 compact, and excluded from the weighted trace, whose first term accounts for it.

6.3. **Compact tails and tightness.**

**Theorem 6.4** (Compact tails)**.** *Let $S \subseteq H^{-2}$ with $\sup_{\varphi\in S} \|\varphi\|_{H^{-2}} \leq B < \infty$.* (i) *For every $\delta > 0$ there is a finite set $K = K(\delta) \subset \mathbb{Z}^2$ with, termwise,*

$$\sum_{k\notin K} w_3(k)\,|\widehat{\varphi}(k)|^2 \leq \delta \sum_{k\notin K} \widetilde{w}_2(k)\,|\widehat{\varphi}(k)|^2 \leq \delta B^2$$

*for every $\varphi \in S$.* (ii) *The closure of $S$ in the carrier is compact. In particular this holds for every sublevel set $\{\varphi : \|\varphi\|_{H^{-2}} \leq R\}$ with $R \geq 0$.*

*Proof. Step 1: the weight comparison.* For $k \neq 0$ the ratio $w_3(k)/\widetilde{w}_2(k) = \|k\|_\infty^{-2}$ tends to 0 along the mode lattice, and that is Proposition 5.4(ii). For $\delta > 0$ put $K := \{k : \|k\|_\infty \leq \lceil \delta^{-1/2} \rceil\}$, a box of finitely many integer vectors containing 0; for $k \notin K$ the ratio is at most $\delta$. So the first inequality of (i) is termwise and the second is (43): a tail bound of $\delta\,\|\varphi\|_{H^{-2}}^2 \leq \delta B^2$ that mentions neither $\varphi$ nor $N$, one finite mode set serving all of $S$. The quantitative form of the same estimate is Proposition 5.4(iii).

*Step 2: a diagonal extraction.* Let $(\varphi_j) \subset S$. For each $k$, $\widetilde{w}_2(k)\,|\widehat{\varphi}_j(k)|^2 \leq B^2$ with $\widetilde{w}_2(k) > 0$, so every coordinate sequence is bounded. Exhaust $\mathbb{Z}^2$ by increasing finite boxes $K_1 \subset K_2 \subset \cdots$. On $K_1$ the coordinates lie in a compact subset of $\mathbb{C}^{K_1}$, so a subsequence converges there; a subsequence of that one converges on $K_2$; and so on. The diagonal subsequence converges at every $k$: the diagonal argument. Let $c = (c_k)$ be the coordinate limit.

*Step 3: convergence in the carrier.* Fix $\delta > 0$ and $K = K(\delta)$. Along the diagonal subsequence,

$$\|\varphi_j - \varphi_{j'}\|_{\mathcal{C}}^2 = \sum_{k\in K} w_3(k)\,|\widehat{\varphi}_j(k) - \widehat{\varphi}_{j'}(k)|^2 + \sum_{k\notin K} w_3(k)\,|\widehat{\varphi}_j(k) - \widehat{\varphi}_{j'}(k)|^2 ,$$

where the first sum tends to 0, being a finite sum of convergent coordinates, and the second is at most $2\delta B^2 + 2\delta B^2 = 4\delta B^2$ by step 1 and $|u-v|^2 \le 2|u|^2 + 2|v|^2$. As $\delta$ is arbitrary the subsequence is Cauchy, and the carrier is complete (Proposition 5.4(i)), so it converges to a point of the closure of $S$. Every sequence in $S$ thus has a subsequence converging in the carrier, so $S$ is totally bounded, and the closure of a totally bounded set in a complete metric space is compact: that is (ii). □

**Theorem 6.5** (Tightness)**.** *Let $L > 0$, $m > 0$, and let $\mu_N$ be the law (20) at $N = n+1$, $a = L/N$. The family $(\mu_N)$ is tight on the carrier: for every $\epsilon > 0$ there is a compact set $K_\epsilon$ in the carrier with $\mu_N(K_\epsilon) \ge 1 - \epsilon$ for every $N$.*

*Proof.* Let $C := C(L, m)$ be (45) and put $T := C/\epsilon + 1$, $K_\epsilon := \overline{\{\varphi : \|\varphi\|^2_{H^{-2}} \le T\}}$, the closure being taken in the carrier; $K_\epsilon$ is compact by Theorem 6.4(ii), applied with $B = \sqrt{T}$. Its complement lies where the energy exceeds $T$, so Markov's inequality applied to the extended-valued energy, which is measurable as a countable sum of continuous functions, with $\mathbb{E}\,\|\varphi_N\|^2_{H^{-2}} \le C$ by Theorem 6.2, gives

$$\mu_N(K_\epsilon^{\mathrm{c}}) \le \frac{1}{T}\,\mathbb{E}_{\mu_N}\,\|\varphi_N\|^2_{H^{-2}} \le \frac{C}{T} < \epsilon,$$

the last inequality because $T > C/\epsilon$. The set $K_\epsilon$ therefore satisfies $\mu_N(K_\epsilon) \ge 1 - \epsilon$, it depends on $\epsilon$, $L$ and $m$ only, and the $+1$ in the cutoff is used exactly here: it makes the Markov tail strictly smaller than $\epsilon$. □

**External theorem 6.6** (Prokhorov)**.** Let $X$ be a Polish space and $(\mu_N)$ a sequence of Borel probability measures on $X$. If $(\mu_N)$ is tight, then there are a strictly increasing sequence $N_1 < N_2 < \cdots$ and a Borel probability measure $\mu$ on $X$ with $\int h\,\mathrm{d}\mu_{N_j} \to \int h\,\mathrm{d}\mu$ for every bounded continuous $h : X \to \mathbb{R}$ [14].

External theorem 6.6 produces a subsequence and nothing about the original sequence: the whole-sequence statement is Theorem 9.1, reached in Section 9 by a separate step and not as a consequence of this theorem. Its Polish hypothesis is met, the carrier being complete (Proposition 5.4(i)) and separable (Theorem 5.6).

One honest limitation closes the section. The only input to Theorem 6.5 is Markov's inequality applied to the second moment of Theorem 6.2: no exponential moment, no layer-cake decomposition and no quantitative modulus enters, so the tail bound is $C/T$ and no better. Consequently this paper asserts no rate of weak convergence, no exponential tail bound and no modulus of continuity in the limit, and a reader who wants one must supply an exponential moment for the family $\mu_N$, which is not proved here. Section 10 records this among the other boundaries of the result.

## 7. Finite Fourier marginals

This section takes the covariance limit of Section 4 from pairings to matrices. Definition 7.1 fixes the finite coordinate maps and records that they are continuous; Theorem 7.2 observes that every finite coordinate law of the lattice sequence is centred Gaussian, and is therefore nothing but its covariance matrix; Lemma 7.3 computes that matrix from the Green pairing of Theorem 4.4; and Theorem 7.4 concludes convergence of the finite-dimensional laws for every finite family. The order of the argument is what needs care, and Lemma 7.3 is where it sits.

### 7.1. Coordinates on the carrier.

**Definition 7.1** (Cylinder maps)**.** Let $r \ge 1$. A *finite family* is a tuple $F = ((k_1, \varepsilon_1), \ldots, (k_r, \varepsilon_r))$ with $k_j \in \mathbb{Z}^2$ and a label $\varepsilon_j \in \{\mathrm{c}, \mathrm{s}\}$ at each slot; *both labels are allowed at every mode, the zero mode included.* Put $\psi_j := \cos_{k_j}$ if $\varepsilon_j = \mathrm{c}$ and $\psi_j := \sin_{k_j}$ if $\varepsilon_j = \mathrm{s}$, with the cosine and sine tests of

Section 6. The *cylinder map* of $F$ is $\Pi_F : \mathcal{C} \to \mathbb{R}^r$,

$$(\Pi_F c)_j := \begin{cases} \Re c_{k_j}, & \varepsilon_j = \mathrm{c}, \\ -\Im c_{k_j}, & \varepsilon_j = \mathrm{s}. \end{cases} \tag{47}$$

The sign in the second line is the one carried by the phase of the lift (19), and it is bookkeeping: for every configuration $\omega$ and every slot $j$,

$$\big(\Pi_F(L_N\omega)\big)_j = \sqrt{w_3(k_j)}\,\omega(\psi_j), \tag{48}$$

because that phase is $e^{-2\pi i\, k\cdot(ax)/L}$, whose real part is the cosine test and whose negative imaginary part is the sine test. So a cylinder coordinate is $\sqrt{w_3(k_j)}$ times the value of the corresponding real test, and $R_F := \operatorname{diag}\big(\sqrt{w_3(k_1)}, \dots, \sqrt{w_3(k_r)}\big)$ is the fixed diagonal scaling from real-test coordinates to cylinder coordinates.

Each coordinate in (47) is a scalar multiple of one of the coordinate functionals of Lemma 5.5, hence continuous, and $\Pi_F$ is continuous into $\mathbb{R}^r$. A continuous map is Borel, so for a Borel probability measure $\mu$ on $\mathcal{C}$ the image $\mu^F := \mu \circ \Pi_F^{-1}$ is a Borel probability measure on $\mathbb{R}^r$, the *$F$-marginal* of $\mu$. The family carrying both labels at each of the modes $k_1, \dots, k_r$ is the complex tuple $(c_{k_1}, \dots, c_{k_r})$ read slot by slot.

The sine test at the zero mode is identically zero, and its slot is retained deliberately rather than discarded as empty: the coordinate it records is $-\Im c_0$, and (47) shows that no other slot sees that coordinate. The family is therefore a determining class, which is what Theorem 8.1 consumes; dropping the slot would leave the cylinder sigma algebra strictly smaller than $\mathcal{B}(\mathcal{C})$ and the characterisation of Theorem 9.1 would fail. The slot costs nothing elsewhere: (51) gives it covariance entry 0 and Theorem 7.2 therefore gives it the degenerate law $\delta_0$, which is the truth, since $(\mathcal{L}_N\omega)(0) = a^2 \sum_x \omega(x)$ is real for every configuration.

### 7.2. **The finite-mode laws and their covariance matrix.**

**Theorem 7.2** (Characteristic functions of the finite-mode laws)**.** *Let $F$ be a finite family with $r$ slots and let $\mu_N$ be the law* (20)*. Then:*

(i) *for every $N$ the marginal $\mu_N^F$ is a centred Gaussian measure on $\mathbb{R}^r$, possibly degenerate, with covariance matrix*

$$\Sigma_{F,N}(j,j') := \sqrt{w_3(k_j)w_3(k_{j'})}\ \operatorname{Re}\mathcal{Q}_N(\psi_j,\psi_{j'}), \tag{49}$$

*and hence with characteristic function $t \mapsto \exp\big(-\frac{1}{2}t^{\mathrm{T}}\Sigma_{F,N}t\big)$;*

(ii) *replacing $Q_N$ in* (49) *by the Green pairing $Q$ of* (22) *gives the matrix*

$$\Sigma_F(j,j') := \sqrt{w_3(k_j)w_3(k_{j'})}\ \operatorname{Re}\mathcal{Q}(\psi_j,\psi_{j'}) = \sqrt{w_3(k_j)w_3(k_{j'})}\ \operatorname{Re} V\left\langle (\psi_j)_L, R(\psi_{j'})_L\right\rangle_E, \tag{50}$$

*which is a limit of the matrices* (49) *entrywise (Lemma 7.3), hence positive semidefinite as an entrywise limit of covariance matrices, hence the covariance matrix of a centred Gaussian measure $\Lambda_F$ on $\mathbb{R}^r$, with characteristic function $t \mapsto \exp\big(-\frac{1}{2}t^{\mathrm{T}}\Sigma_F t\big)$.*

*Both matrices are written down here and identified in Lemma 7.3; this theorem asserts neither that they agree nor that either of them is a limit.*

*Proof.* (i) By Definition 3.4 the physical lattice law $\mu_N^{\mathrm{GJ}}$ is a centred Gaussian measure on $\mathbb{R}^{I_N}$ with covariance $C_N$, and $\omega \mapsto (\omega(\psi_1), \dots, \omega(\psi_m))$ is linear there; a linear image of a centred Gaussian vector is a centred Gaussian vector, and by (48) with (20) that image is $\mu_N^F$. For the entries: by (18) with Lemma 4.3 the second moment $\mathbb{E}_{\mu_N^{\mathrm{GJ}}}[\omega(\psi_j)\omega(\psi_{j'})]$ is $\mathcal{Q}_N(\psi_j,\psi_{j'})$, a real number for real tests (Section 4), and (48) contributes the factor $\sqrt{w_3(k_j)w_3(k_{j'})}$; this is (49). The characteristic function of a centred Gaussian vector with covariance $K$ is $t \mapsto \exp(-\frac{1}{2}t^{\mathrm{T}}Kt)$, the

finite-dimensional Gaussian formula, quoted and not derived. (ii) is a definition, and its second equality is the identification ($Q = R$) proved in Theorem 4.4. □

**Lemma 7.3** (The covariance entries)**.** *In the notation of Theorem 7.2:*

(i) $\Sigma_{F,N} \to \Sigma_F$ *entrywise as* $N \to \infty$*;*

(ii) *the unweighted matrix* $\big(\operatorname{Re}\mathcal{Q}(\psi_j, \psi_{j'})\big)_{j,j'}$ *is the paired cosine/sine periodic-resolvent covariance matrix, and in closed form, for a slot pair drawn from the modes* $k, l$*,*

$$\operatorname{Re}\mathcal{Q}(\cos_k, \cos_l) = \frac{V}{2\lambda(k)}\big(\delta_{k,l} + \delta_{k,-l}\big), \qquad \operatorname{Re}\mathcal{Q}(\sin_k, \sin_l) = \frac{V}{2\lambda(k)}\big(\delta_{k,l} - \delta_{k,-l}\big), \tag{51}$$
$$\operatorname{Re}\mathcal{Q}(\cos_k, \sin_l) = 0;$$

(iii) *for every* $N$ *and every slot,*

$$0 \le \operatorname{Re}\mathcal{Q}_N(\psi_j, \psi_j) \le V/m^2, \qquad \text{and} \qquad \operatorname{Re}\mathcal{Q}(\psi_j, \psi_j) \ge 0,$$

*the second bound being an equality with* $V/\big(2\lambda(k_j)\big)$ *at every slot whose mode is non-zero, with* $V/m^2$ *at the zero-mode cosine and with* $0$ *at the zero-mode sine.*

*Proof.* (i) and the first half of (ii) are one computation, worth isolating because it is where the order of the argument can be reversed.

*Step 1 (the entry is a lattice second moment).* By (48) the coordinate $(\Pi_F c)_j$ is $\sqrt{w_3(k_j)}\,\omega(\psi_j)$, and $\mu_N^{\mathrm{GJ}}$ has mean zero exactly (Definition 3.4), so

$$\Sigma_{F,N}(j,j') = \sqrt{w_3(k_j)w_3(k_{j'})}\,\mathbb{E}_{\mu_N^{\mathrm{GJ}}}\big[\omega(\psi_j)\omega(\psi_{j'})\big],$$

and by (18) with Lemma 4.3 the expectation equals $\mathcal{Q}_N(\psi_j, \psi_{j'})$. This is (49) again, and it shows that the matrix has no source other than the lattice covariance and the weight.

*Step 2 (the limit along a fixed mode pair).* Each $\psi_j$ is a real bounded continuous periodic test, so Theorem 4.4 applies to each pair: its centred-covariance line in (26) gives

$$\mathcal{Q}_N(\psi_j, \psi_{j'}) \longrightarrow \operatorname{Re} V\,\big\langle (\psi_j)_L, R(\psi_{j'})_L \big\rangle_E = \operatorname{Re}\mathcal{Q}(\psi_j, \psi_{j'}),$$

the last equality being the identification ($Q = R$) of its Step 5. Multiplying by the fixed factor $\sqrt{w_3(k_j)w_3(k_{j'})}$ gives $\Sigma_{F,N} \to \Sigma_F$ entrywise, which is (i) and the first half of (ii).

*Step 3 (the entries, as half-sums of exponentials).* Let $e_k(x) := e^{2\pi i\, k\cdot x/L}$, so that $\cos_k = (e_k + e_{-k})/2$ and $\sin_k = (e_k - e_{-k})/(2i)$. The coefficient $f \mapsto F(f)$ of (1) is complex linear in $f$, and at the mode $j$ the coefficient of $e_k$ is exactly $V\delta_{j,k}$: the phase is $e^{2\pi i(k-j)\cdot x/L}$, whose integral over the physical cell vanishes unless $k = j$ and equals $V$ when $k = j$. Inserting the half-sums into (22) leaves only the modes $\pm k$ and $\pm l$ of the two coefficient vectors, and $\overline{F(e_a)} = F(e_a)$ because those coefficients are real; summing the four surviving terms and using $\lambda(-k) = \lambda(k)$ gives the first two identities of (51). For the mixed pair the summand pairs the even coefficient of the cosine with the odd coefficient of the sine, so it is odd under $p \mapsto -p$ and the total vanishes.

*Step 4 (where a reader can go wrong).* Theorem 4.4 already identifies the limit of precisely the pairings of Step 2, and Step 3 evaluates those pairings, so the identification is a short computation and not a new limit theorem: no summability, tail estimate or mode-uniform envelope is spent, because every test is a single mode. The argument can be silently reversed here. The matrix (50) must be obtained from the limit, as in Step 2; postulating it from the symbol $\lambda^{-1}$ while invoking Theorem 4.4 only for the diagonal entries would be a second, unproved assertion, and it is exactly this identification that the machine development carries as a hypothesis on its finite-family theorem rather than deriving from the covariance limit. The two conventions for the sign of the imaginary part in (47) give the same matrix (50): only the mixed entries see that sign, and (51) makes them zero.

*Step 5 (bounds).* For a real test $\psi$, Cauchy–Schwarz in (22), the finite Parseval bound (15) and the bound $\lambda_N \geq m^2$ give

$$\mathcal{Q}_N(\psi,\psi) = V^{-1}\sum_p \frac{|F_N(p)|^2}{\lambda_N(p)} \leq V^{-1}\frac{V^2\,\|\psi\|_\infty^2}{m^2} = \frac{V\,\|\psi\|_\infty^2}{m^2},$$

and the summands are non-negative; with $\|\cos_k\|_\infty \leq 1$ and $\|\sin_k\|_\infty \leq 1$ this is the first claim of (iii). The second is (51) read on the diagonal: it gives $V/(2\lambda(k_j))$ at every slot whose mode is non-zero, $V/m^2$ at the zero-mode cosine, and 0 at the zero-mode sine — the last because $\delta_{0,0}-\delta_{0,-0}=0$, which is the right answer for a slot whose test is identically zero. □

**Theorem 7.4** (Convergence for every finite family)**.** *Let $L>0$, let $m>0$, and let $F$ be a finite family. Then $\mu_N^F \Rightarrow \Lambda_F$ weakly on $\mathbb{R}^r$: for every bounded continuous $\psi:\mathbb{R}^r\to\mathbb{R}$, $\int\psi\,\mathrm{d}\mu_N^F \to \int\psi\,\mathrm{d}\Lambda_F$ as $N\to\infty$.*

*Proof.* Three steps, none of which mentions the carrier.

*(i) The matrices converge.* This is Lemma 7.3(i).

*(ii) The family is tight in $\mathbb{R}^r$, uniformly in $N$.* By Lemma 7.3(iii) each diagonal entry satisfies $0\leq\Sigma_{F,N}(j,j)\leq V/m^2 = L^2/m^2$, so the second moment $\mathbb{E}_{\mu_N^F}|x|^2 = \operatorname{tr}\Sigma_{F,N}$ is at most $rL^2/m^2$ for every $N$. Markov's inequality gives $\mu_N^F\{x:|x|>R\}\leq rL^2/(m^2R^2)\to 0$ as $R\to\infty$, and closed balls in $\mathbb{R}^r$ are compact, so the family is tight. A single Gaussian law on $\mathbb{R}^r$ needs no estimate for this; what the bound supplies is tightness of the whole family at once.

*(iii) The characteristic functions converge.* By Theorem 7.2 the characteristic function of $\mu_N^F$ is $t\mapsto\exp(-\frac12 t^{\mathrm{T}}\Sigma_{F,N}t)$, which by (i) converges for each fixed $t$ to $\exp(-\frac12 t^{\mathrm{T}}\Sigma_F t)$, that of $\Lambda_F$. The limit is continuous at the origin and is the characteristic function of a probability measure, so Levy's continuity theorem — pointwise convergence of characteristic functions to a function continuous at the origin implies weak convergence — gives $\mu_N^F\Rightarrow\Lambda_F$. This is the standard fact that centred Gaussian laws on $\mathbb{R}^r$ with convergent covariance matrices converge weakly [3], and the boundedness of the variances in (ii) is its licence: uniformly bounded diagonal entries give tightness. The limiting covariance may be *degenerate*, and it is: the zero-mode sine slot of Definition 7.1 has limit law $\delta_0$ (Lemma 7.3(iii)). That costs nothing here, because $\Lambda_F$ is a Gaussian probability measure either way and Levy's theorem does not ask for absolute continuity with respect to Lebesgue measure; what the theorem asserts is convergence to $\Lambda_F$, not to a density. □

What Theorem 7.4 establishes is convergence of the pushforwards under every finite coordinate map; by Definition 7.1 these are the laws of the paired cosine/sine coordinates, with the resolvent matrix as limit. What it does not establish is anything about measures on the carrier $\mathcal{C}$. A family of finite-dimensional limits does not by itself produce a measure on $\mathcal{C}$, and convergence of every marginal would not imply weak convergence on $\mathcal{C}$ without a tightness hypothesis; hence the reliance on the uniform moment and compact tails of Section 6 and on the identification of the two sigma algebras in Theorem 5.7. The passage runs: tightness supplies a limit point (Theorem 6.5), continuity of $\Pi_F$ transfers Theorem 7.4 to that limit point, and Theorem 8.1 says two Borel probability measures with the same finite marginals agree. Nothing here produces a measure on the carrier.

*Remark* 7.5 (The index dictionary, once)*.* The family $F$ of Definition 7.1 does not depend on $N$: its modes are integers, fixed once and for all, and only the mesh varies. This is where the two indexing conventions of Hypothesis 2.6 meet, and the dictionary is stated here once, with $N=n+1$ and $a=a_n=L/(n+1)$: the law (20) at index $n$ is $\mu_{n+1}$, and Theorem 7.4 is the statement that these laws converge as $n\to\infty$. Taking the finite family *before* the mesh limit is what makes (50) mesh-free, and it is why the theorem concerns a sequence of laws on one and the same space $\mathbb{R}^r$. By Hypothesis 2.6 the spacing is a function of the index and not a free parameter, so no double limit occurs in this section.

## 8. Cylinder uniqueness and identification

This section supplies the two statements that turn a tight family into one identified limit, and it keeps them separate because they answer different questions. Theorem 8.1 is a statement about the carrier alone: two Borel probability measures there which agree on every finite cylinder set are equal. Lemma 8.2 is a statement about the sequence alone: a weak limit point extracted from $(\mu_N)$ inherits the finite marginals that Theorem 7.4 establishes for the sequence. The first says that the target marginal family admits at most one measure; the second says that every limit point is a measure admitted by it. Together they say that the limit points of the sequence are one point, which is the conclusion Section 9 draws. Neither statement identifies the limit with an operator, and neither is used as if it did.

8.1. **Determination by cylinders.** Throughout, a *cylinder set* is a set of the form $\Pi_F^{-1}(B) \subseteq \mathcal{C}$, where $F$ is a finite tuple of modes, $\Pi_F$ is the finite-mode coordinate map of Definition 7.1 and $B$ is a Borel subset of the finite-dimensional target of $\Pi_F$.

**Theorem 8.1** (Cylinder-to-Borel uniqueness)**.** *Let $\mu$ and $\nu$ be Borel* probability *measures on the carrier $\mathcal{C}$. If $\mu(A) = \nu(A)$ for every cylinder set $A$, then $\mu = \nu$. Equivalently: if all finite-coordinate pushforward laws of $\mu$ and $\nu$ coincide, then $\mu = \nu$.*

*Proof. (i) The cylinders form a $\pi$-system of Borel sets.* Let $A = \Pi_F^{-1}(B)$ and $A' = \Pi_{F'}^{-1}(B')$ be cylinder sets. The map $\Pi_{F\cup F'}$ of the concatenated tuple has $\Pi_F$ and $\Pi_{F'}$ among its coordinate projections, so $A \cap A'$ is the preimage under $\Pi_{F\cup F'}$ of a Borel set, hence a cylinder set. Each cylinder set is Borel, because $\Pi_F$ is continuous (Lemma 5.5); so this family is a $\pi$-system contained in $\mathcal{B}(\mathcal{C})$.

*(ii) It generates the Borel sigma algebra.* Two inclusions. $\subseteq$*:* each $\Pi_F$ is continuous (Lemma 5.5), so $\sigma(\Pi_F) \subseteq \mathcal{B}(\mathcal{C})$. $\supseteq$*:* the tuple $((k, \mathrm{c}))$ is admissible, so $\Re c_k = \Pi_{((k,\mathrm{c}))}$ is $\sigma(\Pi_F)$-measurable; likewise $-\Im c_k = \Pi_{((k,\mathrm{s}))}$, again for every $k$, the zero mode included. Hence each coordinate functional $\pi_k = \Re c_k + i\Im c_k$ is measurable for $\sigma(\Pi_F)$, so $\sigma(\pi_k : k \in \mathbb{Z}^2) \subseteq \sigma(\Pi_F)$, and $\sigma(\pi_k) = \mathcal{B}(\mathcal{C})$ by Theorem 5.7; separability enters through that last step, via Theorem 5.6. The second inclusion is the only place where the definition of Definition 7.1 is used, and it is not decorative. Take the restricted family that keeps the cosine label at every mode and drops the sine label at $k = 0$. The two Dirac measures $\delta_0$ and $\delta_{ie_0}$ on $\mathcal{C}$ — where $e_0$ is the unit vector at the zero mode — agree on every slot of that family ($\Re c_0 = 0$ at both points, and every $c_k = 0$ for $k \neq 0$) and differ in the dropped slot ($-\Im c_0 = 0$ against $-1$). So Theorem 8.1 would fail. The same construction applies to the omission of any single real coordinate: the pair of Dirac masses at two points that differ only in that coordinate separates the restricted family.

*(iii) The Dynkin step.* Fix $\mu$ and put $\mathcal{D} := \{A \in \mathcal{B}(\mathcal{C}) : \mu(A) = \nu(A)\}$. It contains every cylinder set by hypothesis. It contains $\mathcal{C}$ itself, because $\mu(\mathcal{C}) = \nu(\mathcal{C}) = 1$. It is a $\lambda$-system: if $A \subseteq A'$ both lie in $\mathcal{D}$ then $\mu(A' \setminus A) = \mu(A') - \mu(A) = \nu(A') - \nu(A) = \nu(A' \setminus A)$, and if $A_1, A_2, \cdots \in \mathcal{D}$ are pairwise disjoint then countable additivity of both measures gives $\mu(\bigcup_j A_j) = \sum_j \mu(A_j) = \sum_j \nu(A_j) = \nu(\bigcup_j A_j)$. By Dynkin's $\pi$–$\lambda$ theorem a $\lambda$-system containing a $\pi$-system contains the sigma algebra that $\pi$-system generates, so by (i) and (ii) $\mathcal{D} = \mathcal{B}(\mathcal{C})$, that is $\mu = \nu$.

*(iv) Where the word probability is used, and what the conclusion buys.* Steps (ii) and (iii) hold for any two finite measures of equal total mass; it is the hypothesis that both are probability measures which makes that total mass equal, and the hypothesis cannot be dropped. For two infinite measures agreeing on every cylinder set, the difference in Step (iii) is undefined at the first complement, and equality of total mass is exactly what is missing. From $\mu = \nu$ we obtain equality of integrals of every bounded Borel function, and every bounded continuous function $\mathcal{C} \to \mathbb{R}$ is Borel because it is continuous. The sigma algebra equality (ii) is what makes "agrees on every cylinder" and "has the same law" the same requirement. □

8.2. **A subsequential limit inherits the target marginals.** The second statement is the only place where the marginal theorem of Section 7 meets the extraction theorem of Section 6. It says nothing about how many limit points there are.

**Lemma 8.2** (A subsequential limit inherits the target marginals)**.** *Let $L > 0$, let $m > 0$, let $(\mu_N)$ be the sequence* (20) *of finite-source Fourier pushforwards, let $(N_j)$ be a strictly increasing sequence of indices and suppose that $\mu_{N_j} \Rightarrow \mu$ weakly on $\mathcal{C}$. Then for every finite tuple $F$ of modes the pushforward $(\Pi_F)_*\mu$ is the resolvent Gaussian law $\Lambda_F$ that Theorem 7.4 records as the limit of the finite-mode laws $(\Pi_F)_*\mu_N$.*

*Proof.* Fix $F$, and abbreviate $\mu_N^F := (\Pi_F)_*\mu_N$ and $\mu^F := (\Pi_F)_*\mu$. The map $\Pi_F$ is continuous (Lemma 5.5), and a continuous image of a weakly convergent sequence of Borel probability measures converges weakly to the image of the limit: for every bounded continuous $\psi$ on the finite-dimensional target, the composition $\psi \circ \Pi_F$ is bounded and continuous on $\mathcal{C}$, so

$$\int \psi \, \mathrm{d}\mu_{N_j}^F = \int (\psi \circ \Pi_F) \, \mathrm{d}\mu_{N_j} \longrightarrow \int (\psi \circ \Pi_F) \, \mathrm{d}\mu = \int \psi \, \mathrm{d}\mu^F.$$

Hence $\mu_{N_j}^F \Rightarrow \mu^F$. For the second limit, note what $\mu_N^F$ is: the stored coordinate at a mode is $\sqrt{w_3}$ times the Fourier coordinate, so its law is the image of the law of the real cosine and sine coefficients of that mode under the fixed reassembly map which multiplies by $\sqrt{w_3}$ (Definition 7.1). By Theorem 7.4 the laws of those real coefficients converge weakly, for every finite family of test functions, to the resolvent Gaussian law; a fixed continuous reassembly carries that convergence to $\Lambda_F$, and a subsequence of a convergent sequence has the same limit. Hence $\mu_{N_j}^F \Rightarrow \Lambda_F$. The target of $\Pi_F$ is a metric space, so its Borel probability measures with the weak topology form a Hausdorff space, in which a sequence has at most one limit; therefore $\mu^F = \Lambda_F$. □

These two results are the last probabilistic steps of Part I. What stands between them and the theorem is purely topological: at this point one knows that every subsequence of $(\mu_N)$ has a further subsequence converging to one and the same measure, and the passage from that statement to convergence of the whole sequence mentions no measure and no probability at all. Section 9 makes that passage, as Step 5 of the proof of Theorem 9.1, and names the external fact it uses (external theorem 9.2).

## 9. The main theorem

The content of this section is the order in which the ingredients are applied. Let $L > 0$ and $m > 0$, let $(\mu_N)$ be the sequence (20) of finite-source Fourier pushforwards, indexed as in Hypothesis 2.6 by $N = n + 1$ with $a = L/(n + 1)$, so that $aN = L$ at every index, and let $\mathcal{C}$ be the carrier of Definition 5.1.

**Theorem 9.1** (Main theorem)**.** *Let $L > 0$ and $m > 0$.*

(i) Existence. *There is a Borel probability measure $\mu_{\mathrm{free}}$ on $\mathcal{C}$ such that $\mu_N \Rightarrow \mu_{\mathrm{free}}$ weakly on $\mathcal{C}$, for the whole sequence and not only along a subsequence.*

(ii) Marginals. *For every finite tuple $F$ of modes the pushforward $(\Pi_F)_*\mu_{\mathrm{free}}$ is the resolvent Gaussian law $\Lambda_F$ of Theorem 7.4: the paired cosine/sine Gaussian law whose covariance entries are the periodic-resolvent pairings identified in Theorem 4.4 and Lemma 7.3.*

(iii) Characterisation. *$\mu_{\mathrm{free}}$ is the only Borel probability measure on $\mathcal{C}$ all of whose finite-mode marginals are the laws $\Lambda_F$.*

*Proof. Step 1 (a relatively compact family).* Each $\mu_N$ is a Borel probability measure on $\mathcal{C}$ (Eq. (20), Lemma 5.8), so the family to which the compactness theorem is applied lives where it must. By Theorem 6.2 the expected *square* of the $H^{-2}$ norm of $\mu_N$ is bounded by the constant $C(L, m)$ of Lemma 6.1, uniformly in $N$; Theorem 6.5 converts that second moment of the norm, through

Markov's inequality and the compactness of Theorem 6.4, into tightness of the family $\{\mu_N\}$ in $\mathcal{C}$. Tightness is the hypothesis under which external theorem 6.6 turns that family into a relatively compact one, in the weak topology of Remark 5.10.

*Step 2 (extraction along any subsequence).* The carrier is a complete separable metric space (Theorem 5.6), so external theorem 6.6 applies: for every strictly increasing sequence $(N_j)$ of indices there are a strictly increasing subsequence $(N_{j_l})$ of $(N_j)$ and a Borel probability measure $\mu$ on $\mathcal{C}$ with $\mu_{N_{j_l}} \Rightarrow \mu$. Nothing is claimed yet about which measure $\mu$ is.

*Step 3 (that limit has the target marginals).* By Lemma 8.2, $(\Pi_F)_*\mu = \Lambda_F$ for every finite tuple $F$ of modes.

*Step 4 (that limit is the only one admissible).* Let $\mu$ and $\mu'$ be two measures obtained as in Steps 2 and 3, possibly from different starting subsequences. By Step 3 both have marginals $\Lambda_F$, hence $\mu$ and $\mu'$ agree on every cylinder set, and Theorem 8.1 gives $\mu = \mu'$. The cylinder families of Definition 7.1 include the sine slot at the zero mode, whose test is identically zero and whose coordinate is $-\Im c_0$; that slot is what makes the families separate every coordinate of the carrier, and without it the cylinder sigma algebra would be strictly smaller than $\mathcal{B}(\mathcal{C})$ and this step would not close. This is the characterisation of (iii): Theorem 8.1 says that the family $\Lambda_F$ admits at most one Borel probability measure, and Lemma 8.2 says that every extracted limit is such a measure. Statement (ii) is Step 3 applied to one such limit.

*Step 5 (from subsequences to the whole sequence).* Extract one limit $\mu_{\text{free}}$ from the identity subsequence as in Step 2; by Step 4 the choice of extraction does not matter. Let $(N_j)$ be any strictly increasing sequence of indices. By Steps 2 to 4 the sequence $(\mu_{N_j})$ has a further subsequence converging to $\mu_{\text{free}}$, which is the hypothesis of external theorem 9.2; its conclusion is $\mu_N \Rightarrow \mu_{\text{free}}$ for the whole sequence, that is (i). □

**External theorem 9.2** (Subsequence criterion for whole-sequence convergence)**.** Let $(x_N)$ be a sequence in a topological space $X$ and $x \in X$. Suppose that every strictly increasing sequence of indices has a further strictly increasing subsequence $(N_l)$ along which $x_{N_l} \to x$. Then $x_N \to x$.

The criterion is the standard subsequence characterisation of convergence, and its proof is two lines; it needs no metric space. If $x_N$ did not converge to $x$, an open neighbourhood $U$ of $x$ would be missed infinitely often; the indices where it is missed form a subsequence with no term in $U$, and by hypothesis that subsequence has a further subsequence converging to $x$, hence eventually inside $U$, a contradiction. This is the only use of external theorem 9.2 in the paper.

**Corollary 9.3** (Uniqueness)**.** *Let $(N_j)$ be a subsequence of the index sequence and let $\nu$ be a Borel probability measure on $\mathcal{C}$ with $\mu_{N_j} \Rightarrow \nu$. Then $\nu = \mu_{\text{free}}$.*

*Proof.* By Theorem 9.1(i) the subsequence converges to $\mu_{\text{free}}$, so $\nu$ and $\mu_{\text{free}}$ are two limits of one sequence in the space of Borel probability measures on a metric space, which is Hausdorff. Hence they coincide: this is uniqueness of the limit, not the characterisation of it. □

Four things are deliberately not claimed, and each is a boundary of this paper rather than a weakness of the proof; Section 10 develops them.

- *No identity beyond the law of the Fourier coefficients.* Theorem 9.1 concerns the push-forwards $\mu_N$, and Lemma 5.8 identifies them as the laws of a Borel carrier-valued map of the lattice field, so the convergence proved here *is* a statement about the lattice field laws in exactly that sense. What is not asserted is any statement about a total lift specified independently of $\mathcal{L}_N$ — in particular the series-defined total lift of the development, whose measurability is an input there (Remark 5.9) — and no statement about the field as a distribution on $T_L^2$ rather than as its sequence of Fourier coefficients.
- *No covariance operator on the carrier.* What is proved is that the finite marginals of $\mu_{\text{free}}$ are the target laws of Theorem 7.4, the paired cosine/sine Gaussian laws whose

characteristic functions are computed in Theorem 7.2. The resolvent of Lemma 4.1 is constructed on the unit-mass $L^2$ space of the torus, not on $\mathcal{C}$, and no covariance operator is constructed on $\mathcal{C}$ here. The distinction is between a construction and a consequence: saying that $\mu_{\text{free}}$ *is* the Gaussian measure with covariance operator $(m^2 - \Delta)^{-1}$ as an object on the carrier requires that operator on the carrier, which is not built; saying which measure it is requires only Theorem 8.1, and that is what is proved. What this paper proves is the marginal statement, not the operator statement (Section 10).

- *No rate.* No rate of convergence, no exponential tail and no quantitative modulus is asserted. The tightness of Step 1 is Markov's inequality applied to one second moment of the norm, which is sufficient for external theorem 6.6 and supplies nothing quantitative beyond it.
- *Free field only.* The theory constructed is the free (Gaussian) one. Neither an interacting ultraviolet limit nor an Osterwalder–Schrader reconstruction is proved, and Theorem 9.1 asserts nothing about the interacting model.

## 10. Scope, boundaries, and the wider programme

Let $L > 0$ and $m > 0$. Theorem 9.1 proves that the finite-source Fourier pushforwards $\mu_N$ of the fixed-volume lattice Gaussian laws converge weakly, as a whole sequence, to the single Borel probability measure on the carrier $\mathcal{C} = H^{-3}$ that their finite Fourier marginals determine. Its hypotheses are proved in the same chain and not assumed: the tightness of the family (Theorem 6.5), with the uniform $H^{-2}$ moment behind it, and the limits of the finite-marginal laws (Theorem 7.4), whose target is the paired cosine and sine Gaussian law of the periodic resolvent. Separately, Theorem 4.4 proves the analytic limit: the covariance integrals of the physical lattice laws $\mu_N^{\text{GJ}}$ converge to the pairing with the resolvent of Lemma 4.1. This section does the opposite job. It records what neither statement asserts, places what they do assert among the branches of the programme, and points to the machine-checked layer.

10.1. **What the paper does not assert.** Five boundary statements belong in the body rather than in a footnote: a reader who supplies one of them from context would read the paper as claiming more than it proves. Each carries the reason it is a boundary of the statement, not a gap in its proof.

(i) *What converges is the law of the Fourier coefficients.* The sequence in Theorem 9.1 is $\mu_N = (L_N)_* \mu_N^{\text{GJ}}$ of Definition 3.6, and Lemma 5.8 identifies this as the law of a Borel carrier-valued map of the lattice Gaussian field. So the theorem does concern the lattice field laws; what it does not touch is any total lift specified independently of $L_N$ — in particular the series-defined lift of the development, which carries its own measurability input (Remark 5.9) — and it says nothing about the field as a distribution on $T_L^2$. "The lattice field laws converge" is therefore asserted in the sense of the law of the coefficient sequence, which is the sense in which $\mu_N$ is a measure on $\mathcal{C}$ at all.

(ii) *No identification with a distribution-defined Sobolev space, and no Laplacian domain theorem.* The carrier is the explicit weighted $\ell^2$ space of Definition 5.1, and Proposition 5.4 is a statement about it: isometrically $\ell^2$, with $H^{-2}$ sitting inside continuously. The paper does not assert that $\mathcal{C}$ agrees with a Sobolev space of distributions defined independently of it, and it proves no strong domain theorem for an unbounded Laplacian. Every estimate here is a weighted sum over modes, and no step is taken over from such an identification.

(iii) *No covariance operator on the carrier, and no operator-level Gaussianity.* The resolvent $R$ of Lemma 4.1 is constructed on the $L^2$ space of the torus, not on $\mathcal{C}$, and no covariance operator is constructed on $\mathcal{C}$ here. What is proved is the marginal statement of Theorem 9.1(ii): for each finite tuple $F$ of modes, $(\Pi_F)_* \mu_{\text{free}}$ is the paired cosine and sine Gaussian law of Theorem 7.4. Read as an operator statement, the proposition that $\mu_{\text{free}}$ is the Gaussian

measure whose covariance operator is $(m^2 - \Delta)^{-1}$ is therefore not asserted. A reader who wants it must either construct the operator on the carrier or use cylinder determination, since the cylinders generate the Borel sigma algebra (Theorem 5.7) and the marginals determine the measure (Theorem 8.1). It is listed here because it is a step this paper does not take.

(iv) *No rate, no exponential tail, no modulus of continuity.* Tightness (Theorem 6.5) is Markov's inequality applied to one second moment of the norm, the bound of Theorem 6.2, plus compactness of $H^{-2}$ sublevel sets in the weaker topology (Theorem 6.4). No exponential moment and no layer-cake argument enters, so no rate in $N$, no exponential bound on tail probabilities and no modulus of continuity for $\mu_{\text{free}}$ is asserted. A second moment of the norm is all external theorem 6.6 consumes through tightness, and nothing more.

(v) *The measure limit is two-dimensional.* The mode lattice $\mathbb{Z}^2$, the weights $w_2, w_3$, the cylinder maps, the carrier and the law sequence are all two-dimensional, so Theorem 9.1 is a theorem about the two-dimensional carrier, and no dimensional generalisation of that carrier is claimed (Remark 2.9). The covariance limit is the one statement whose source and machine-checked versions are registered for a general dimension $d$; Part I writes the two-dimensional case that Section 2 fixes. Nothing here says what becomes of the weights, the trace constant or the compact tail at $d \neq 2$.

10.2. **Where the result sits in the programme.** This paper is one step of one branch of a programme, and three other branches of the same project are outside it. Each is named so that a reader knows what this paper is not about; none is evaluated, and nothing is promised.

- *The free scalar theory in four dimensions* is the free massive Euclidean field on $\mathbb{R}^4$: the Gaussian measure on a space of distributions, its covariance, the Osterwalder–Schrader axioms for it, and the physics reconstructed from them. This paper does not touch it; nothing here is stated on $\mathbb{R}^4$ or at infinite volume.
- *The interacting two-dimensional theory* is the $P(\varphi)_2$ model [7]: interacting finite-volume laws, their ultraviolet limit at fixed volume, the axiom verification, and the mass gap. This paper does not touch it; the laws whose covariance limit is Theorem 4.4 are Gaussian at every mesh size.
- *The QED programme* is the non-perturbative construction of four-dimensional quantum electrodynamics, with its reconstruction and its physical content. This paper does not touch it; no gauge-theoretic object occurs in the argument.

The atlas of Fig. 2 is a module-level dependency graph of the wider Lean development of the project: its nodes are modules of that development as a whole, not of the package this paper reports, and its edges are declaration dependencies. The package that Part II reports is one cluster inside the atlas, and the layers tabulated in Section 11 are the interior of that cluster.

10.3. **Relation to the literature.** 16 works are cited in this paper, in three capacities that must not be confused. Four support a standard fact where it is used – the Riesz representation behind Lemma 4.1, the Prokhorov extraction of external theorem 6.6, the convergence of Gaussian laws with convergent covariance matrices in Theorem 7.4, and the definition of weak convergence on a metric space in Remark 5.10 – and each is cited at that point rather than here. Three are the closest relatives of what this paper does, and they are compared in the paragraphs after the list rather than disposed of in a line: [4] for the convergence statement, [2, 13] for the recipe and the topology, and [5] for the machine-checked layer. The remaining nine support nothing: they are cited for position, and each of the following relations is a position, not a dependency. Eight of them are named below; the ninth, [7], is named among the exclusions above.

- [4] prove that the scaling limit of the discrete Gaussian free field on a random graph approximating a compact Riemannian manifold is the manifold continuum Gaussian free

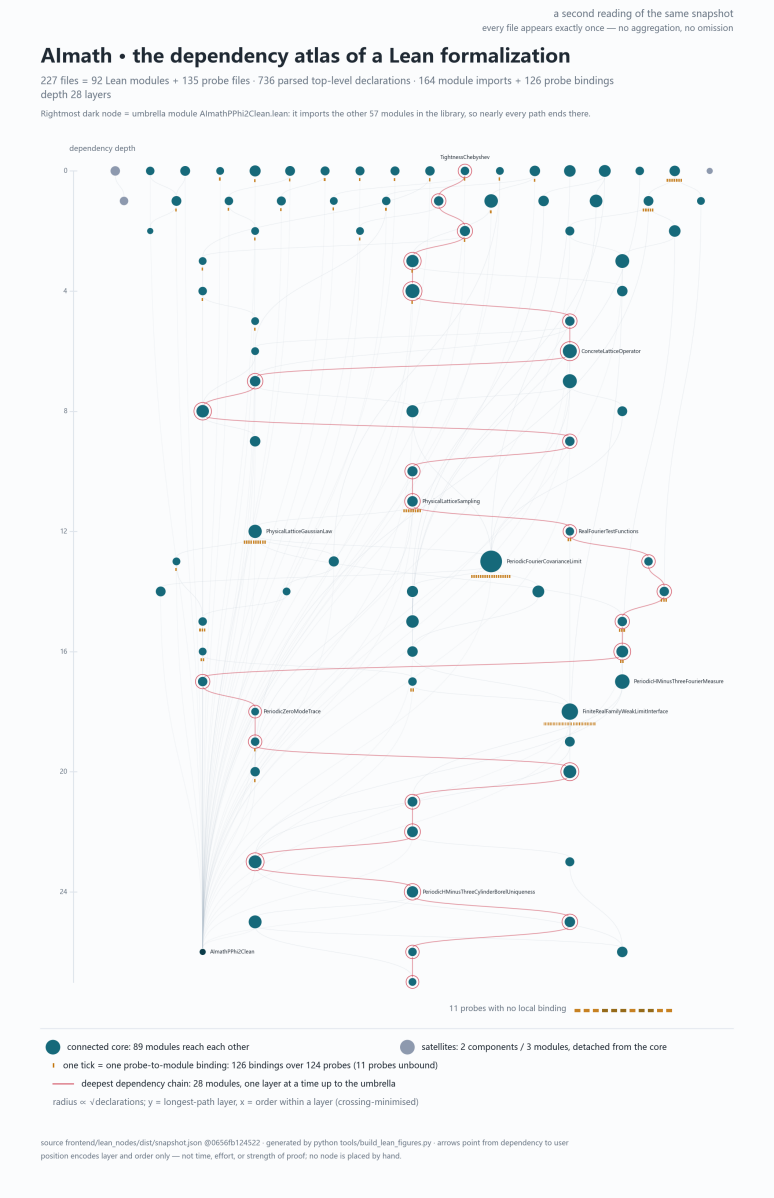



FIGURE 2. Dependency atlas of the Lean development. Each node is a module and each edge a declaration dependency. The atlas is a snapshot of a slightly wider tree than the package of Section 11: it reports 92 modules and 135 probe files against the package's 89 and 132, and the figure's own legend identifies the three extra modules as satellites detached from the core rather than as part of it. The dense core the atlas names is the package. The surrounding directory of Section 11 is much larger than either count — it holds 43 further modules — and none of it is drawn here. The figure is therefore best read as a picture of where the reported chain sits inside the tree that was measured, not as evidence about anything outside that chain.

field, in law and in the strong Sobolev topology after a Voronoi tessellation. It is the closest published relative of the convergence statement here, and it is compared below.

- [2] is the standard account of the discrete Gaussian free field in two dimensions, and of what a scaling limit of it is; it is also where the recipe this paper follows is set out in words

– tightness, then a subsequential limit, then a unique identification of the limit object. [13] obtain convergence of nonlinear functionals in $H^{-s}(D)$ for random conductances. Both are compared below, because together they fix the standard of comparison for a convergence claim in a negative Sobolev space.

- [5] report a Lean 4 formalisation of the free massive Euclidean field in four dimensions together with the Glimm–Jaffe axioms, and record that the Minlos, nuclearity and Goursat inputs of its first release were subsequently proved or avoided. It is the nearest machine-checked relative of Part II, and it is compared below.
- [11, 12] state the axioms: the properties (analyticity, reflection positivity, Euclidean invariance, clustering and regularity) that a Euclidean measure must satisfy before reconstruction can produce a Wightman theory. They fix the target of the programme, and none of those properties is verified here.
- [6] is the standard account of the programme as a whole, and so the reference point against which its order of steps is usually read; no step here is taken from it.
- [15] and [8] are the monograph and the paper-length accounts of the two-dimensional model, and so the reference point for what a completed construction of that model contains; neither is used here.
- [10] is the reference point for the lattice Gaussian layer, the free Euclidean field and its Markov property, which is the layer the laws of Definition 3.4 belong to.
- [9] is the nuclearity and Minlos route from a positive-definite functional to a measure on the dual of a nuclear test space, which is the route this paper does not take: Section 5 builds the carrier as an explicit weighted sequence space, and external theorem 6.6 is the only extraction principle used.

The comparison that matters most is with [4], because both papers take a discrete Gaussian field to its continuum counterpart in a Sobolev topology, and both do it by pushing the discrete object into a function space. Two of the differences below do not favour this paper, and are stated first for that reason. Their convergence holds in the *strong* topology of $H^{-s}$; ours is weak convergence of probability measures on a carrier that happens to be an $H^{-3}$-type sequence space, and a strong-$H^{-s}$ statement is the stronger one. And their model is a random graph on a manifold, which is a harder index set than a periodic square lattice. What this paper adds is elsewhere: an explicit carrier with no Minlos step, an identification of the limit by its finite Fourier marginals alone, convergence of the whole sequence rather than of a subsequence, and a machine-checked counterpart. Four differences, each of which states what this paper does and none of which states anything about theirs. *Model:* they use a random graph, because a compact manifold has no canonical grid; here the index set is the deterministic periodic square lattice at fixed physical volume $aN = L$, and the operator is the massive $-\Delta_a + m^2$ with the Glimm–Jaffe normalisation of Definition 3.4. *Identification of the limit:* they identify their limit as the continuum Gaussian free field on the manifold; here the limit is identified by its finite Fourier marginals and by nothing else (Theorem 8.1), so no density against a reference Gaussian is ever constructed. *Topology and mode of convergence:* theirs is convergence in law in the strong Sobolev topology; here it is weak convergence of Borel probability measures on the carrier, and of the whole sequence rather than of a subsequence (Theorem 9.1(i), Corollary 9.3). *Evidence:* Part II of this paper is a machine-checked counterpart of its own statement, at the resolution Section 13 records.

The recipe deserves a sentence of its own, because it is not this paper's. [2] sets out the pattern a scaling limit of this kind is proved by – establish tightness, extract a subsequential limit, identify the limit object uniquely – and every step of Theorem 9.1 has a counterpart there. What differs is the third step: the identification here runs through Theorem 8.1 (two measures that agree on every finite Fourier marginal coincide), whereas the standard route identifies through characteristic functionals or through the Osterwalder–Schrader axioms. The topology is also weaker than the standard output: [13] obtains $H^{-s}$ convergence for nonlinear functionals, and a negative Sobolev

norm is where this literature lives. In short, what this paper adds is the fixed-volume massive lattice covariance limit, a carrier constructed without Minlos (Section 5), a whole-sequence tightness-to-uniqueness chain, and a formal counterpart of the same statement – and none of those four is a new method.

The machine-checked layer has a nearest relative too, and it is a larger project. [5] formalises the free massive Euclidean field in four dimensions and the Glimm–Jaffe axioms in Lean 4, using the Schwartz-space route, and its release notes record that the Minlos, nuclearity and Goursat inputs of the first release were afterwards proved or avoided. Three things follow, and the third is the one a reader should carry away. First, "a machine-checked lattice-to-continuum limit" is not new, and Part II must not be read as claiming it. Second, "no Minlos" is not by itself a contribution: Minlos has been formalised, and can also be avoided, in that project. Third, what this paper's Part II does contribute is a different route to a smaller target — an explicit weighted sequence carrier, a whole-sequence limit, and a footprint of `propext`, `Classical.choice`, `Quot.sound` for each reported theorem without a project axiom anywhere in the chain — together with an evidence chain in which every number and every digest can be recomputed from the delivered sources.

Nothing above is a statement about priority: whether a comparable theorem exists elsewhere is no longer left as a gesture, since the closest ones are cited and compared, but a claim of priority would need a search this paper has not done and does not make.

Part II reports the machine-checked layer of the paper: 10 declarations, from the covariance limit through the whole-sequence limit, with Section 11 mapping each onto its module. They are not uniformly the same statements as Part I, and where they differ the difference is in the direction of weaker hypotheses or weaker conclusions, never stronger: the finite-family declaration consumes as a hypothesis the entrywise covariance identification that Lemma 7.3 proves here, so what Lean closes is the passage from that identification to weak convergence, not the identification itself. How far coverage reaches is a separate question with a separate answer, and Section 13 is where it is answered.

## Part 2. The Machine Verification

### 11. The Lean development: organisation and module map

The formalisation is a separate Lake package, `AImathPPhi2Clean`, whose sources live under `formal/lean/s6/pphi2_clean/`, built against Mathlib on Lean v4.34.0-rc2. `lake-manifest.json` pins the Mathlib commit `85e3a25e`, which is the revision of the tag `lakefile.lean` names. It has 89 modules and 13505 lines of Lean, together with 132 audit probe files; a comment-stripped count, computed at build time by the paper's own script and never typed here, finds 0 `sorry` and 0 project axioms in those modules. The library target builds: `lakebuildAImathPPhi2Clean` elaborates all 89 modules with no error on the pinned revision, including the 8 that the root file does not reach (Section 13). The package sits inside a larger development: the surrounding directory `formal/lean/s6/` holds 43 further modules with 10730 lines, which this paper does not take as its subject, does not count, and does not claim to cover. Every declaration quoted below lives in the namespace `Pphi2Clean.Continuum`.

This is not the first machine-checked development of a lattice-to-continuum limit, and the paper should not be read as saying that it is. Section 10 compares it with one that formalises the free massive field in four dimensions and the Glimm–Jaffe axioms [5]; the differences that matter are the carrier, the mode of convergence, and the fact that no project axiom appears in any footprint here.

The modules are organised in layers, and the layers follow the order of the conclusions rather than the order of the proofs: each one exposes an interface that the next one consumes, so the map below is also a reading order. The entry module in the right-hand column is the module to open first for that layer.

| Layer | What it does | Entry module(s) |
|---|---|---|
| (a) periodic torus $L^2/H^1$ and the discrete operator | Periodic $H^1$ as a closed value-gradient graph completion; the finite periodic lattice, its discrete Laplacian with spacing $a$, and an eigenbasis of $Q = -\Delta_a + m^2$ | `PeriodicTorusH1`, `ConcreteLatticeOperator`, `PeriodicFourierLaplacian` |
| (b) the massive resolvent and its energy solver | The weak massive resolvent from an energy form and a Riesz adjoint; sampled and continuum Fourier coefficients; the cutoff tail uniform in the mesh; the three covariance limits restated with the resolvent as target | `PeriodicFourierCovarianceLimit`, `PeriodicTorusResolventLimit` |
| (c) physical lattice sampling and the Gaussian law | Sampling of a test at physical sites with the cell volume as weight; the normalised finite covariance; zero mean, characteristic function, mixed-moment identity, and the covariance-to-weak-convergence transfer | `PhysicalLatticeGaussianLaw`, `PhysicalLatticeSampling` |
| (d) negative-Sobolev carriers, topology and Borel structure | The $H^{-2}$ and $H^{-3}$ weights, the completion as $\ell^2$ with its weight and deweight maps, the compact-tail comparison, the coordinate embeddings, second countability, separability, and the cylinder sigma algebra equal to the Borel one | `PeriodicHMinusThreeCompletion`, `PeriodicHMinusThreeTopologyBorel` |
| (e) uniform $H^{-2}$ moment and $H^{-3}$ tightness | The trace of the squared $H^{-2}$ norm under the finite Gaussian source and its mesh-uniform bound; the extended energy, its compact sublevels, the Markov tail, and tightness of the law sequence | `PeriodicHMinusTwoUniformTightness`, `PeriodicHMinusThreeFixedVolumeTightness` |
| (f) finite real families, characteristic functions and covariance convergence | The resolvent covariance matrix, its positive semidefiniteness and Gram factor, the canonical target Gaussian, and unconditional weak convergence for every finite family of tests | `FiniteRealFamilyWeakLimitInterface`, `FiniteFourierCoordinateConvergence` |
| (g) cylinder convergence | All the lift maps, the finite-source measure, the paired cosine and sine cylinder family, and convergence of every finite Fourier marginal | `PeriodicHMinusThreeCylinderConvergence`, `PeriodicHMinusThreeFourierMeasure` |
| (h) Prokhorov subsequence | Tightness in the form Prokhorov consumes, and a strictly monotone subsequence converging to a Borel probability measure on the carrier | `PeriodicHMinusThreeProkhorovSubsequence` |
| (i) cylinder-to-Borel uniqueness | Finite-coordinate cylinders as a $\pi$-system, the Borel-generation step discharged from separability, and equality of two Borel probability measures with equal indexed cylinder marginals | `PeriodicHMinusThreeCylinderBorelUniqueness` |
| (j) final assembly | Inheritance of the target marginals by any subsequential limit, existence and uniqueness of the whole-sequence limit, and the packaged existence-with-cylinders statement | `PeriodicHMinusThreeWeakLimitMarginal` |

Layers (a) and (b) are analysis on the torus, layers (c) to (e) are the lattice laws together with the carrier and the single moment, layers (f) to (i) are the convergence and identification steps, and layer (j) applies them in order.

The correspondence with Part I is by result, not by proof step. Every name in the right-hand column lives in the namespace `Pphi2Clean.Continuum` and the namespace is omitted there. Appendix A lists the 10 declarations frozen in the two statement contracts, with the module that

declares each of them; the remaining names of the table below are declarations of the same package whose role is described here rather than frozen, and their existence is checked against the Lean sources by the paper's own gate rather than against the registry.

| Result in Part I | Declaration(s) in `Pphi2Clean.Continuum` |
|---|---|
| Theorem 4.4 | `fixedVolumeCovarianceInner_tendsto_torusFourierGreenPairing`<br>`physicalGJMeasure_fixedVolume_mixedMoment_tendsto_torusFourierGreenPairing`<br>`physicalGJMeasure_fixedVolume_covariance_tendsto_torusFourierGreenPairing` |
| Theorem 5.6 | `periodicHMinusThreeCompletion_separable` |
| Theorem 5.7 | `periodicHMinusThree_cylinder_sigma_eq_borel_of_polish` |
| Theorem 6.5 | `periodicHMinusThree_fixedVolume_freeField_tight` |
| Theorem 7.4 | `physicalFamilyEvalProbabilityMeasure_fixedVolume_weak_tendsto_of_periodic_fourier_canonical` |
| Theorem 8.1 | `periodicHMinusThree_probabilityMeasure_ext_of_indexed_cylinder_marginals` |
| Lemma 8.2 | `periodicHMinusThree_weak_limit_has_expected_cylinder_marginal`<br>`periodicHMinusThree_fixedVolume_subsequential_limit_has_expected_cylinder` |
| Corollary 9.3 | `periodicHMinusThree_fixedVolume_weak_limit_unique` |
| Theorem 9.1 | `periodicHMinusThree_fixedVolume_weak_limit_exists_with_expected_cylinders`<br>`periodicHMinusThree_fixedVolume_weak_limit_exists`<br>`periodicHMinusThree_fixedVolume_weak_limit_has_expected_cylinder` |

For the 10 declarations of the chain the correspondence is statement by statement: each of the nine results in the left-hand column has at least one declaration on its right whose conclusion is that statement, and the three limits inside Theorem 4.4 are the three declarations of the first row, one per line of that theorem. Two rows carry a qualification, and they are the two the reader is most likely to overread. The finite-family declaration consumes the entrywise covariance identification as a hypothesis — that is, the very statement Lemma 7.3 derives from the covariance limit — so what the machine closes there is the passage from that identification to weak convergence of the finite families, with the target supplied by the Fourier-constructed factor. The covariance input is exactly what Theorem 7.4 does not assume, because Part I spends Section 4 proving it. The cylinder-Borel declaration is stated under a Polishness input rather than unconditionally, and the same module derives that input from separability of the carrier (`periodicHMinusThreeCompletion_separable`), which is how the chain that consumes it discharges it; Part I's version is unconditional because Section 5 proves separability before stating it. Neither qualification makes a row claim more than Part I does. That is the precise sense in which Part II mirrors Part I, and it is what the table lets a reader check without reading Lean: the paper's statement on the left, the declaration that carries it in the middle column, and the module that declares it in Appendix A. The row for Corollary 9.3 is the weakest entry in the table and must be read as such: the declaration it names says that two limits of one filter coincide, which is Hausdorff uniqueness of a filter limit, and not that the limit is characterised. The characterisation is the cylinder rows of the table, as it is in Part I. What the table does not say is anything about the three parts of Theorem 9.1 separately: existence and the marginal statement are carried by the three declarations of the last row, while part (iii), the uniqueness of the limit, is carried by the cylinder-uniqueness row above it — that row's declaration is the formal counterpart of Part I's part (iii), though Part I states no Lean name.

Two qualifications belong here rather than in a footnote, because a reader who misses either will overread the table. First, the formal development is a separate unit from this paper's prose: it has its own module structure, its own definitions and its own conventions for normalisation, so the package is not the paper transcribed, and the paper does not claim that the Lean code follows the proof of Part I line by line. Statement identity is what is on offer here, and it is all that is on offer. Second, coverage is not uniform across the layers, and Section 13 collects the five places where the development diverges from a maximally favourable reading of it: which modules the

package root reaches, a final-theorem-shaped wrapper that assumes its own conclusion, the total-lift measurability input that is not proved, the two-dimensional restriction of the measure chain while the covariance theorems are stated for general dimension, and what a complete build does and does not establish.

Part II's two appendices are ordered accordingly. Appendix A (Appendix A) is the declaration ledger and Appendix B (Appendix B) the axiom-footprint transcript, and both are generated on every build from the statement registry and from the recorded transcript rather than typed, because a hand-maintained list of declaration names is exactly the kind of object that goes stale without anyone noticing.

## 12. Statement contracts and the axiom firewall

Two devices carry the discipline of the formal side, and neither of them is a proof: a *statement contract*, which fixes what is to be proved, and an *axiom firewall*, which bounds what the proof of it may assume. This section describes both, and separates what each establishes from what it leaves open; the divergences that remain are the subject of Section 13.

A statement contract is an entry of the statement registry of the development, `formal/lean/challenges/_registry.yaml`. For one node of the development it freezes three things: the modules of the delivered package that declare the results, the theorem names in fully qualified form, and a digest of each statement, held beside the entry in a per-node file. The registry carries 14 contracts. The two this paper reports are `PPHI2_FIXED_VOLUME_COVARIANCE_LIMIT` and `PPHI2_FIXED_VOLUME_FREE_FIELD_MEASURE_LIMIT`. The first freezes the covariance declarations, among them `fixedVolumeCovarianceInner_tendsto_torusFourierGreenPairing`; the second freezes the measure-limit declarations, among them `periodicHMinusThree_fixedVolume_weak_limit_exists`. "Fully qualified" means that the namespace is part of the frozen string: each of the 10 names this paper reports lives in `Pphi2Clean.Continuum`, and the ledger of Appendix A omits that prefix only because the omission is itself performed mechanically.

12.1. **What freezing buys.** The digest is taken of what Lean prints for `#check @` of a name, that is, of the elaborated type with all implicit and instance binders, not of the source text as typed. A later strengthening, weakening or reparametrisation of a statement therefore produces a different digest. The runner reports the mismatch and refuses the statement; it does not overwrite the recorded value, so the contract stays red across runs, and the only way past it is to record the new statement together with a written reason. That is what freezing buys: a statement cannot change quietly, and a change accepted once cannot become invisible afterwards.

12.2. **Both contracts are remote-only.** Both entries are marked remote-only and name the separate Lake project that hosts the recorded digests, `/home/zhengyj/AImath-lean/AImathPPhi2Clean`. The runner reads that field before it does anything else, and reports `SKIP` with the declared project as the reason. **A `SKIP` is not a pass.** It records that the digest was neither confirmed nor refuted locally, because the recorded values live beside that project; a reader who takes the absence of a failure for a success reads the output backwards. The flag is about *where the digest runner points*, not about whether the package can be elaborated: the package is in this repository and, as Section 11 and the reproduction section record, it builds in place. The evidence standing behind these two contracts is therefore two-part: the digests and the registry entry on the project side, and, on this side, the transcript of Appendix B, produced by building the modules that declare the 10 theorems and elaborating one audit file against them.

12.3. **The axiom firewall.** The permitted set of the development is 3 names, the axioms of the Lean kernel and of `Mathlib`:

`propext`, `Classical.choice`, `Quot.sound`.

Every declaration this paper reports has been checked with `#print axioms`; each contract's check file issues that command once per frozen name, next to the `#check` that produces the digest, so one file exercises both devices. This paper reports 10 declarations, and the footprint of every one of them is equal to the permitted set: no axiom introduced by the project appears, and no placeholder standing for an omitted proof appears. Two thresholds are in force and they are not the same one. The contract runner refuses a footprint that contains any name outside the permitted set. The generator of Appendix B applies two further rules of its own: it refuses a transcript that is empty, that is not tab-separated, whose rows are not all equal to one another, or any of whose rows leaves the permitted set. The equality with the permitted set *as a set* is enforced one step later, by the paper's own gate, which is why a footprint that merely stayed inside the permitted set would not reach this paper.

12.4. **The negative control.** A test that has never failed has not been shown to be able to fail. The package therefore ships a probe, `audit/negative_axiom.lean`, 3 lines long: it injects `axiom injectedAxiom : False`, derives `theorem polluted : False := injectedAxiom`, and prints the footprint of the derived theorem, and one line of its run output records a footprint carrying the injected axiom and none of the permitted three. What that establishes is that the footprint test has a case in which it must fail, so a green run of it is not an artefact of a check that can only say yes. What it does not establish is anything about any particular declaration: the probe is a separate file, it is not a perturbation of the frozen names, its recorded output is a bare measured line rather than a transcribed session with commands and a date, and it exercises the footprint test only, not the digest. The control is therefore attested textually – the file is in the tree and its output line is committed beside it – and not by a transcript of the standard of Appendix B.

12.5. **Two roles: dependence and intent.** A footprint is a statement about dependence, and dependence is not content. Equality with the permitted set says that the proof term of the declaration uses no axiom outside that set; it says nothing about whether the statement is the intended one, because the footprint is read off the proof term and the shape of the statement is not part of what it records. A declaration that proves a weaker proposition, or a different one, has the same footprint as a declaration that proves the intended proposition. What addresses intent is the other two devices. The digest of the contract detects a change in a statement over time, and the ledger of Appendix A fixes, for every reported declaration, its module and its role in the argument. The division of labour is therefore: the firewall reports what was assumed, the digest reports what was frozen, the ledger reports where the declaration sits. All three are necessary, none is sufficient, and what the three together still leave unchecked is stated in Section 14.

## 13. What is machine-checked and what is not

The formal side of this paper is easier to overstate than to state, so the divergences between the sources and a maximally favourable reading are collected here as numbered facts. Each is a property of the sources, not a hedge on a result: the declarations asserted below are the declarations that exist, and the sentences below say how far the evidence for them extends. The influence of each item on the mathematics is none; the influence on what a reader may conclude is total.

(i) *Reachability.* 89 modules exist in the package, and the package root aggregator imports a subset of them directly. 8 modules are not reachable from the root file even transitively. Two of them are load-bearing: `PeriodicHMinusThreeWeakLimitMarginal`, which declares 6 of the 7 registered measure-limit theorems, and `PeriodicHMinusThree ProkhorovSubsequence`, which declares the Prokhorov extraction step of the argument. Consequently a build of the *root file* does not evidence the axiom footprints of those declarations: nothing reachable from it contains them, so a claim that the root covers the development is a claim about a subset of the package. What does evidence them is a build

of the library target, which elaborates every module including the unreachable ones, and that is the build Appendix B records. The distinction matters to a reader who builds the aggregator and stops: the result is green and says nothing about 8 modules, 6 of whose declarations are on this paper's chain.

(ii) *A vacuous wrapper.* The development contains a final-theorem-shaped interface, `lattice FreeFieldMeasure_tendsto_periodicFreeGaussian`, whose proof is `exact hfinal`: it assumes its own conclusion. This paper does not use it, and no result on these pages depends on it. It exists as a contract-shaped interface, listing the data that a completed assembly would have to supply – tightness, the finite marginals, the cylinder-to-Borel input and the subsequential identification – and a reader who found it by its name and its shape would take it for the paper's theorem. It is not, and citing it as such would be filing an empty citation.

(iii) *The formalisation's lift is not the paper's.* The development's carrier-valued lift is a total function, defined coordinatewise with the series appearing only in the $\ell^2$ side condition, together with the convention that it is 0 off the convergence set; its measurability is one conjunct of an input of the formal development, the other conjunct asserting almost-everywhere agreement with the raw weighted lift. That convention is *weaker* than a range statement and must not be read as one: the function is total at the price of the convention, not of a theorem about it. Nothing in the mathematics turns on it. The paper's carrier-valued representative is $L_N$ of Definition 3.6, whose total range, Borel measurability and entry formula are proved outright (Lemma 5.8), and the chain of Section 9 runs through $L_N$, not through the development's lift; on the finite-source route the development moreover proves the two pushforwards equal without using the input. So a reader comparing the two sources should read the development's input as a fact about one implementation choice, not as a gap in Theorem 9.1. What is genuinely absent on both sides is stated in Section 10.

(iv) *Dimension.* The measure chain is hard-wired to two dimensions: the carrier, the Fourier index set, the cylinder maps and the law sequence are all built over the two-dimensional index set, and no parameter for the dimension occurs in them. The registered covariance theorems are stated for a general dimension $d$. The two statements are therefore not the same statement generalised differently: the covariance statements are *registered* for general $d$ (that is how the Lean declarations quantify), this paper writes them in $d = 2$, and the measure limit, as formalized, is available in $d = 2$ only. No dimension other than two is claimed anywhere in Part I (Remark 2.9).

(v) *What a green build proves.* A complete build proves the Lean declarations that the sources state, under the hypotheses that the sources attach to them, on the toolchain recorded in the transcript; a complete build of this package exists and is recorded in Section 14. It proves nothing about the mathematics of a hypothesis: a hypothesis is an input, and a build that consumes it is not evidence for it. It proves nothing about statements outside the library target of Item 1. And the count of 0 occurrences of `sorry` is a statement about the text of the sources, obtained by a scan that strips comments, rather than a statement about the mathematics: a source file free of placeholders can still formalize a proposition weaker than the one a reader has in mind, which is why Item 3 and Item 4 are stated here rather than absorbed into the count. The mathematical boundary of this paper – the coverage comparison between these ten declarations and the rest of the development, and the absence of a rate – is a different boundary, and Section 10 draws it.

## 14. Reproduction

Two entry points are supported and they are not equivalent. The paper's own gate is `python build_all.py`, run from the paper directory. In order it regenerates the structural counts and both

appendices – the declaration ledger from the sources, the axiom transcript from the recorded engine output – runs the negative controls of the paper checks and then the checks themselves, compiles with Tectonic 0.17.0, refuses a PDF older than any source, and stamps the digest of every source and of the output into `build/LAST_BUILD.txt`. The four checks are: that every Lean name quoted in the body is a name the generated ledger lists, or else a declaration of the package found in the Lean sources; that the large structural counts in the body are macros rather than typed values; that labels, cross-references, citations and bibliography keys close and every section carries a label; and that the axiom story is self-consistent, in the sense of Section 12. Stated precisely: the count check rejects a typed literal only when the macro it should have used has at least three digits or a value of at least forty, so a hand-typed `two` would pass it, and the discipline for small numbers is editorial rather than mechanical; the check also fails when a macro that the build computes is quoted nowhere, which is the failure mode a typed count leaves behind. The reference check covers every reference command the document uses, `\eqref` included. The compile step is followed by a scan of the log: a build that reports success while leaving an overfull box or an undefined reference is a failed build here, because the engine treats both as warnings and the reader sees both as defects. The project around the paper carries 137 proof documents, 147 verification probes and 23 failure cards; none is shipped with this paper and none is an input to any claim above. The second entry point is the bare compile, `tectonic -C main.tex`, which builds the PDF from whatever the generated files currently contain. It is what a reader who wants only the document runs, and it proves nothing about the sources: a stale generated file compiles as happily as a fresh one.

One property of the build is worth stating because a digest of the output depends on it. Tectonic stamps the PDF with the wall clock, so two compiles of unchanged sources used to produce two different files, and every recorded digest of the PDF went stale for a reason that had nothing to do with the paper. `build_all.py` therefore pins `SOURCE_DATE_EPOCH`, and the build is byte-reproducible: compiling the same sources twice yields the same SHA-256, which is what makes the digest in `build/LAST_BUILD.txt` a property of the sources rather than of the afternoon.

The footprints are reproduced in the separate project, in two steps, using the recorded commands. First the declaring module is built, which populates the object files that the second step reads. Then a short file is written whose imports are the modules under audit and whose body is one `#print axioms <name>` line per declaration, and that file is elaborated with `lake env lean`. The toolchain is Lean v4.34.0-rc2, the same one the transcript header records. This is exactly the command whose output `ext/axiom_transcript.txt` transcribes. The transcript was regenerated on 2026-11-03 by `audit/make_transcript.py`, on the sources in this repository, and it is the artefact to compare against. The digests of the statement contracts are in a different position and must not be confused with the transcripts: they are recorded values in the per-node files held beside the registry entries described in Section 12, and because both contracts are marked remote-only the runner reports `SKIP` before it computes a digest at all. That flag governs the digest runner, which points at the project holding the recorded values; it does not say that the package cannot be elaborated here, and Section 11 and the paragraph below record that it can. Nothing shipped with this paper exercises the digest, but a reader can run the build.

A reproducer should expect the following, and should treat any deviation as a finding. The package builds on the revision it pins — Lean v4.34.0-rc2 with Mathlib `85e3a25e`, the commit `lake-manifest.json` pins for the tag `lakefile.lean` names — and the toolchain it reports is the one the transcript header records: the footprints are valid on that toolchain, and comparing them across toolchains is not a comparison. Each `#print axioms` line prints the permitted set — all 10 of them do, in Appendix B; a line carrying a name outside it, or an empty footprint in place of a set, would be a finding, and the appendix contains none. The paper's gate exits zero. And the compiled document carries no unresolved reference or citation: two independent mechanisms enforce that, since the third check refuses a document whose cross-references, citation keys or section labels do not close, and the log scan refuses an output on which the engine reported one.

One thing is deliberately not automated, and it is the limit of the audit. The correspondence between a sentence of this paper and a Lean statement is checked at the level of declaration names and of the frozen digests, never at the level of the statement's meaning: the name check confirms that a quoted name exists in the delivered development, and the digest confirms that the statement behind a frozen name has not changed since it was accepted, but no script compares the prose of a theorem environment with the type of a declaration, so a sentence that misdescribes the statement it cites would pass every check we run. That residual gap is why Section 13 exists as a list of facts rather than as a claim of completeness.

One episode belongs in this section because it bears on how much a reproducer should trust the apparatus. An audit of this paper, carried out after it was first assembled, found that one module of the package was missing the header of a declaration — the body survived, the signature did not — so the sources as delivered could not have been elaborated, and every claim resting on a build would have been unreproducible. The header is restored, and the sources have since been migrated to the Mathlib revision the toolchain pins; the whole package now elaborates, on that revision, with no error. The transcript in Appendix B was *regenerated* afterwards, by `audit/make_transcript.py`, from the Lean run on these sources: it therefore records a command that was executed on the delivered tree, not a transcription of an earlier session. The script also writes the digest of each declaring module into the transcript header, so a change to any of them invalidates the record visibly rather than silently.

What that does not buy is stated in the same breath. A footprint is evidence about the proof term of one declaration; it is not evidence that the declaration says what the corresponding theorem of Part I says, and it is not evidence about the mathematics of any hypothesis in the statement. Those two gaps are the subject of Section 13 and of the paragraph above, and neither is closed by a green build.

## Appendix A. Declaration ledger

Every declaration frozen in the two statement contracts is listed below with its fully qualified Lean 4 name, the module of the delivered package that declares it, and its role in the argument. The names are reproduced mechanically from the statement registry of the development, so that the reader can locate each claim in the sources without transcribing an underscore. Declarations that the body quotes but that are not frozen in a contract are named in the correspondence table of the Lean overview section, which is where their role is described; this appendix does not claim to list them, and the paper's gate checks them against the Lean sources.

### PPHI2_FIXED_VOLUME_COVARIANCE_LIMIT

| Declaration | Module, and its role in the argument |
|---|---|
| `fixedVolumeCovarianceInner_tendsto_torusFourierGreenPairing` | `PeriodicFourierCovarianceLimit` fixed-volume convergence of the lattice covariance pairing to the torus Green pairing |
| `physicalGJMeasure_fixedVolume_mixedMoment_tendsto_torusFourierGreenPairing` | `PeriodicFourierCovarianceLimit` the same limit for the mixed second moment of the physical lattice law |
| `physicalGJMeasure_fixedVolume_covariance_tendsto_torusFourierGreenPairing` | `PeriodicFourierCovarianceLimit` the same limit for the centred covariance; this is the input the weak-limit theorem consumes |

### PPHI2_FIXED_VOLUME_FREE_FIELD_MEASURE_LIMIT

| Declaration | Module, and its role in the argument |
|---|---|
| periodicHMinusThreeCompletion_separable | PeriodicHMinusThreeTopologyBorel<br>the carrier is separable, and complete by the completion; separable and complete gives Polish, which is what carries the Prokhorov machinery |
| periodicHMinusThree_probabilityMeasure_ext_of_indexed_cylinder_marginals | PeriodicHMinusThreeWeakLimitMarginal<br>two Borel probability measures agreeing on every indexed cylinder agree |
| periodicHMinusThree_fixedVolume_subsequential_limit_has_expected_cylinder | PeriodicHMinusThreeWeakLimitMarginal<br>the finite-mode limits pass to any subsequential limit extracted by Prokhorov |
| periodicHMinusThree_fixedVolume_weak_limit_exists | PeriodicHMinusThreeWeakLimitMarginal<br>existence of a weak limit point of the whole sequence |
| periodicHMinusThree_fixedVolume_weak_limit_has_expected_cylinder | PeriodicHMinusThreeWeakLimitMarginal<br>that limit point has the prescribed finite Fourier marginals |
| periodicHMinusThree_fixedVolume_weak_limit_unique | PeriodicHMinusThreeWeakLimitMarginal<br>any two limit points coincide, so the limit is independent of the subsequence |
| periodicHMinusThree_fixedVolume_weak_limit_exists_with_expected_cylinders | PeriodicHMinusThreeWeakLimitMarginal<br>the main theorem: whole-sequence weak convergence to the uniquely determined law |

The prefix Pphi2Clean.Continuum is common to all ten names and is omitted in the table.

## Appendix B. Axiom-footprint transcript

The transcript below was produced by #print axioms inside the delivered Lake project, after the library target was built: the command lake build AImathPPhi2Clean elaborates every module of the package, including the ones the root aggregator does not reach. The commands and the engine output are in ext/axiom_transcript.txt of the source tree; nothing here is paraphrased.

| Declaration (prefix Pphi2Clean.Continuum omitted) | Axiom footprint |
|---|---|
| fixedVolumeCovarianceInner_tendsto_torusFourierGreenPairing | propext, Classical.choice, Quot.sound |
| physicalGJMeasure_fixedVolume_mixedMoment_tendsto_torusFourierGreenPairing | propext, Classical.choice, Quot.sound |
| physicalGJMeasure_fixedVolume_covariance_tendsto_torusFourierGreenPairing | propext, Classical.choice, Quot.sound |
| periodicHMinusThreeCompletion_separable | propext, Classical.choice, Quot.sound |
| periodicHMinusThree_probabilityMeasure_ext_of_indexed_cylinder_marginals | propext, Classical.choice, Quot.sound |
| periodicHMinusThree_fixedVolume_subsequential_limit_has_expected_cylinder | propext, Classical.choice, Quot.sound |
| periodicHMinusThree_fixedVolume_weak_limit_exists | propext, Classical.choice, Quot.sound |
| periodicHMinusThree_fixedVolume_weak_limit_has_expected_cylinder | propext, Classical.choice, Quot.sound |
| periodicHMinusThree_fixedVolume_weak_limit_unique | propext, Classical.choice, Quot.sound |
| periodicHMinusThree_fixedVolume_weak_limit_exists_with_expected_cylinders | propext, Classical.choice, Quot.sound |

The permitted set of the development is Classical.choice, Quot.sound, propext, i.e. exactly the three axioms of the Lean kernel and Mathlib. In particular no declaration above depends on an axiom introduced by the project, and none depends on sorryAx.

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